The Error of Polynomial Voxels for Elastic Waves

Isolated Exactly on a Layer of Cubes, and Known before the Solve in Three Dimensions

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Abstract

Volume-integral (coupled-dipole) methods divide a heterogeneous elastic medium into cubic voxels coupled through the Green's tensor of a reference medium. Each voxel represents the medium within it and the wavefield within it, each in a basis of its own, and we determine the error that a given pair of bases leaves, in a form known before the solve. On a layer of voxels, which for cubes is the layer itself so that its exact solution isolates the discretisation error, a voxel with the wavefield to Legendre degree p and the contrast to degree r converges at order $\min(2p+2, 2r+2)$, with leading error $c_p(kd)^{2p+2}$ in closed form ($c_0 = 1/12$, $c_1 = 1/720$, $c_2 = 1/100800$, k the wavenumber and d the voxel side), for incident P, SV and SH waves at normal and oblique incidence, in a uniform background and in a plane-stratified one. At second order in the contrast the error is the relative projection error of the medium, the fraction of the squared norm of its profile that polynomials of degree $\min(p, r)$ on the cells cannot represent, so the two bases must be matched. In three dimensions the error of the scattered far field is the sum of two terms, both available before the solve: the error of the term of first order in the contrast, from one pass over the cells and in closed form for a cell of uniform contrast, with leading term $-(\mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}})\, d^2/12$ for a constant field; and the nonlinear fraction of the response times the relative projection error of the medium, times a factor of order one that we derive in closed form, the nonlinear fraction itself following from two products with the matrix of a coarse grid, with no solve. A sphere whose contrast falls smoothly to zero converges at the orders the layer predicts, and against its exact solution the second term has one constant, for constant and degree-one cells, on every grid. Every result is derived symbolically and checked against an independent implementation or an exact solution.

1 Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper serves is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides

the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T-matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and strain inside the voxel, which the calculation must find. Each is expanded in a basis on the cell, and the two bases are separate choices. Here both are Legendre polynomials, the wavefield to degree p and the contrast to degree r; the usual scheme, a uniform field in a cell of constant properties, is $p = r = 0$. Neither choice is new in itself. In higher-order volume integral methods for electromagnetics the material within an element is represented by polynomials whose orders are independent of those of the field [1], as it is in finite elements, and voxels that carry a linear field are in use [2]. What has been missing is the law that relates the two bases: the order a given pair attains, and the error it leaves. The answer decides the cost of everything built on it, because a voxel that represents its contents more accurately can be larger, and the number of voxels, not the size of each one's state, sets the cost of the multiple-scattering solve.

Why the question is hard to answer. For the media such a scheme is built for, there is no exact solution. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green's tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the single voxel's response is matched to cancel it, refinement converges to the wrong answer without revealing it.

Why a layer matters beyond a test. A plane-stratified medium is more than a convenient test of a voxel scheme. Seismic media are dominated by their layering, and a volume-integral scheme becomes competitive when its reference medium is itself stratified: the Green's tensor of the layered background then carries the layering exactly, with its reflections, conversions and guided waves, and the voxels need represent only the departure of the medium from it. That is the setting of the cell-based scheme of Nestor [3]. The law established here on a layer of voxels holds in such a background too: the orders, the term of first order and the projection-error law survive a background of two half-spaces and one of three layers, at normal and at oblique incidence (§8, Appendix C).

The method: a layer of voxels. This paper therefore answers the question in the simplest configuration in which a voxel scheme runs in full (every voxel, every coupling, the whole multiple-scattering series, at normal and at oblique incidence) and the exact answer is nevertheless known: a layer filled with voxels. Because cubes tile space, a layer of identical voxels does not approximate a homogeneous layer; it *is* one. Any difference between its reflection and transmission and those

of the continuous layer is therefore error of the discretisation alone, with no modelling error and no shape error behind which to hide. A sphere, the usual test object for volume-integral and discrete-dipole methods [4], cannot give this: its error mixes the discretisation with the staircase approximation of a curved surface by cubes. The layer plays the part of the patch test in finite elements: reproducing a uniform medium correctly is necessary, though not sufficient, for a scheme to be trusted in a heterogeneous one. The experiment is then made harder in three steps, each with an exact solution of its own: a randomly stratified layer, in which every cell is homogeneous but the cells differ; a layer whose properties vary smoothly within every cell, which is where the basis for the medium is tested; and a sphere whose contrast falls smoothly to zero at its surface, which carries the layer's prediction into three dimensions without a staircase. The coupling and the single site that the layer validates are the same in any medium; a laterally heterogeneous medium remains to be tested.

The answer. (1) *The order is set by the weaker basis.* The scheme converges at order $\min(2p+2, 2r+2)$ in the cell size (§8). A contrast held constant in each cell therefore caps every voxel at second order in a medium that varies within the cells, however rich the basis of its field; and a contrast of higher degree than the field is wasted. (2) *The leading error is in closed form, for every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$ with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§7); the orders 2, 4 and 6 of $p = 0$, 1 and 2 hold for P, SV and SH waves at normal and oblique incidence. For the collocation scheme, the uniform-field voxel in its simplest form, the error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§6), and the opposite signs explain why no correction of the single site can improve both (§13). (3) *The error that remains is the single site's, and it is the projection error of the medium.* The single-site T-matrix sums the rescattering within a voxel to all orders and the coupling between voxels is exact, but the T-matrix is that of the voxel as represented, its contrast projected to degree r and its field held to degree p. The error of the term of second order in the contrast equals the *relative projection error* of the medium, the fraction of the squared norm of its profile that polynomials of degree $\min(p,r)$ on the cells cannot represent (Eq. (23)): exactly in the long-wave limit, and to within 1.3% for all nine pairs with $p, r \le 2$ (§8). The wavefield basis must therefore carry what the medium imprints on the field, not only the wave, and the two bases must be matched. (4) *The error is known in advance.* The projection error is a property of the medium and the grid alone, so the cell size and the degree that a required accuracy needs are chosen before anything is solved. (5) *Where the medium is piecewise constant on the cells the projection error vanishes*, a constant suffices for the medium, and the degree of the field alone sets the order: in a randomly stratified model one first-moment voxel per model layer reaches a reflection accuracy of 10^{-6}, where the uniform-field voxel needs about forty (§8), and in the uniform layer it allows voxels 58 times larger in each direction (Appendix H). (6) *In three dimensions* the smoothly graded sphere converges at second order with the mean-only voxel, as the layer predicts (§10). And the error of a three-dimensional body separates into two terms, both known before the solve: the error of the term of first order, in closed form for a cell of uniform contrast, and the nonlinear fraction of the response times the relative projection error of the medium, times a factor of order one derived in closed form (§§11–12).

What makes the experiment exact. The answer above can be trusted only if nothing but the two bases is approximated. Three results, set against the coupled-dipole method described below, secure that. (a) *An exact coupling.* The integrated Green tensor and the filtered coupled dipoles approximate the field of a cell; for cubes the source-cell average is exact, because the cube's form factor vanishes on the reciprocal lattice (the tiling identity, §5), and at normal incidence the lattice kernel is in closed form with no lattice sum. (b) *No polarisability to choose.* The polarisability, a modelling choice in the coupled-dipole method, is here a derived object: the moments of the Green's tensor over the cube, taken as distributions, are in closed form [5], and the self term the single site carries is the one the averaged kernel omits, so the discrete system is a collocation of the continuum equation (§5). (c) *The elastic state.* A voxel carries a force (through its density contrast) and a moment (through its stiffness contrast), so its state has nine components, displacement and strain, the coupling is a 9×9 tensor, and the single site's self-interaction is the elastic depolarisation of a cube, the discrete counterpart of the inclusion tensor of Eshelby [6]. Cubes serve all three: they tile space, so a uniform region cut into cubes is still that region, and their regular lattice makes the coupling a convolution, solved by fast transforms.

The coupled-dipole method. Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green's function is the multiple-scattering description of Foldy [7] and Lax [8]. In electromagnetics it became a numerical method as the coupled-dipole method, or discrete dipole approximation (DDA): each cell of a cubic grid is a polarisable point dipole driven by the incident field and by the fields of every other dipole [9, 10]. The method is a collocation of the volume integral equation for the electric field, with the cell's self-interaction absorbed into its polarisability [11], and Yurkin and Hoekstra [12] review it in that form. Its accuracy turns on two ingredients, and this paper takes up both for elasticity. The first is the polarisability. The Clausius–Mossotti value is the static response of a sphere; the lattice dispersion relation of Draine and Goodman [13] adjusts it so that an infinite lattice propagates waves as the continuum does to a given order in kd; Rahmani et al. [14] instead derive it from the exact static solution for the whole body, so that the method is exact in the long-wavelength limit. The second is the coupling. Integrating the Green tensor over the source cell rather than sampling it at the centre improves the accuracy, most of all for strong contrasts [15], and the filtered coupled dipoles of Piller and Martin [16], revived by Yurkin et al. [17] for large refractive indices, replace the point dipole by a source of finite extent whose spectrum is band-limited to the grid. The coupling summed over a lattice extends the method to periodic structures [18]. Its error converges at second order in the dipole spacing when shape errors are absent [4], and for a cube, the one shape a cubic grid represents without a staircase, Yurkin and Kahnert [19] reach relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the mathematical side, Costabel et al. [20] show that the method is a finite section of an infinite block Toeplitz matrix, stable for a large range of the contrast but not for all of it, and Costabel [21] that in a one-dimensional model it converges at second order in the interior, with a boundary layer in which the consistency error does not vanish. The layer studied here is the elastic counterpart of that cube: a body the grid represents exactly, so that the discretisation error can be isolated. What this paper adds is the closed form: for cells that tile space, the coupling that makes the

lattice reproduce the continuum exactly, and the order and the constant of the error that remains, as functions of the two bases. In the terms used here the coupled-dipole method is the pair $p = r = 0$, and its refinements improve the polarisability and the coupling while leaving both bases constant in the cell.

Its elastic counterparts. The elastic problem has the same volume integral equation [22], and every cell-based solver of it is a coupled-dipole method whether or not it carries the name. Nestor [3] divided a two-and-a-half-dimensional medium into cells that interact through the background Green's tensor, each carrying a contrast renormalised by its own near field, with the multiple-scattering series summed. Kanaun and Levin [23] discretise the inclusion problem with Gaussian approximating functions whose matrix elements are analytic. Yang et al. [24] solve a stress–velocity form of the integral equation on a grid by conjugate gradients with fast Fourier transforms, and Touhei [25] does the same for a half space with a generalised Fourier transform. Jakobsen [26] and Jakobsen et al. [27] build the seismic forward and inverse problems on the transition operator of the discretised Lippmann–Schwinger equation, the first in the acoustic approximation and the second in arbitrary anisotropic elastic media with a nine-component state vector.

Error estimation for adaptive cells. Polynomial cells, and adaptive trees of them, are not new. For the scalar Lippmann–Schwinger equation in two dimensions, Ambikasaran et al. [28] discretise on an adaptive quad tree with a polynomial on every leaf, splitting a leaf until it holds a set number of points per wavelength and resolves the contrast to a tolerance, and Gopal and Martinsson [29] accelerate the direct solve. In three dimensions Malhotra and Biros [30, 31] represent volume sources by polynomials of bounded total degree on the leaves of an adaptive octree, refined where the highest coefficients exceed a tolerance, and evaluate their potentials with a kernel-independent fast multipole method [32] for the Poisson, Stokes and low-frequency Helmholtz problems, at up to 10^{10} unknowns. Polynomials on the cells of a dyadic tree are the multiwavelets of Alpert [33], in which integral operators are sparse [34]. In electromagnetics, large volume elements carry fields of high polynomial order [35], with the material represented to an order of its own [1], and voxels with a degree-one field are solved with fast Fourier transforms [2]. Where error is estimated for such methods it is estimated *a posteriori*, from a computed solution that is then refined and solved again [36]. For elastic waves the volume integral equation is solved on uniform grids at low order [37], and fast multipole methods exist for boundary integral equations [38, 39].

This paper. It extends the formulation of Nestor [3] to cubic voxels and is in two parts. Part I establishes the law on a layer. §2 sets out the discrete medium and §3 the continuous layer it must reproduce; §4 shows why the point coupling fails, and §5 why the source-averaged coupling succeeds exactly. §§6–7 take up the wavefield basis and §8 the medium basis and its projection error. Part II carries the law to three dimensions: the Galerkin cells (§9), the graded sphere (§10), the error in two terms (§11) and the matching of the bases (§12). §13 discusses what the results imply, §14 states what is established, and §15 concludes. Every result is derived symbolically in

Mathematica and checked against an independent Python/Fortran implementation or an exact solution.

Part I

The law on a layer

2 Setting

The discrete medium. Space is tiled by cubes of side d (half-side $h = d/2$). Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with Lamé parameters λ, μ, density ρ, P and S speeds α, β and P-wave modulus $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates are ordered (z, x, y) with z down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y,\ \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \tag{1}$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the 9×9 single-site T-matrix maps the exciting state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source voxel's force and moment to the state at voxel i. The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j)\, \mathbf{T}_j\, \psi_j, \tag{2}$$

with ψ^0 the incident state.

Where the kernel comes from. Write the medium as the background plus the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} + \rho\omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde}\, \varepsilon_{de}$ and summation over repeated indices. Splitting $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta\lambda\, \delta_{bc}\delta_{de} + \Delta\mu\, (\delta_{bd}\delta_{ce} + \delta_{be}\delta_{cd})$ the contrast stiffness, and moving the contrast terms to the right leaves the background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho\, u_b + \partial_c m_{bc}, \qquad m_{bc} = \delta c_{bcde}\, \varepsilon_{de}, \tag{3}$$

a force from the density contrast, and the divergence of the contrast stress m, the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the background, the displacement is $u = u^0 + \int \mathbf{G} f \, \mathrm{d}\mathbf{y}$, and the moment term integrates by parts,

$$\int G_{ab}(\mathbf{x} - \mathbf{y})\, \frac{\partial m_{bc}}{\partial y_c}\, \mathrm{d}\mathbf{y} = -\int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x} - \mathbf{y})\, m_{bc}\, \mathrm{d}\mathbf{y} = +\int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y})\, m_{bc}\, \mathrm{d}\mathbf{y}, \tag{4}$$

the boundary term vanishing because m is confined to the scatterers, and the last step holding because $\mathbf{G}$ depends only on the separation. The moment therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when the moment is the contrast stress itself. With the moment tensor defined instead through its equivalent body force $-\partial_c[M_{bc}\, \delta(\mathbf{x} - \mathbf{y})]$, as in Aki and Richards

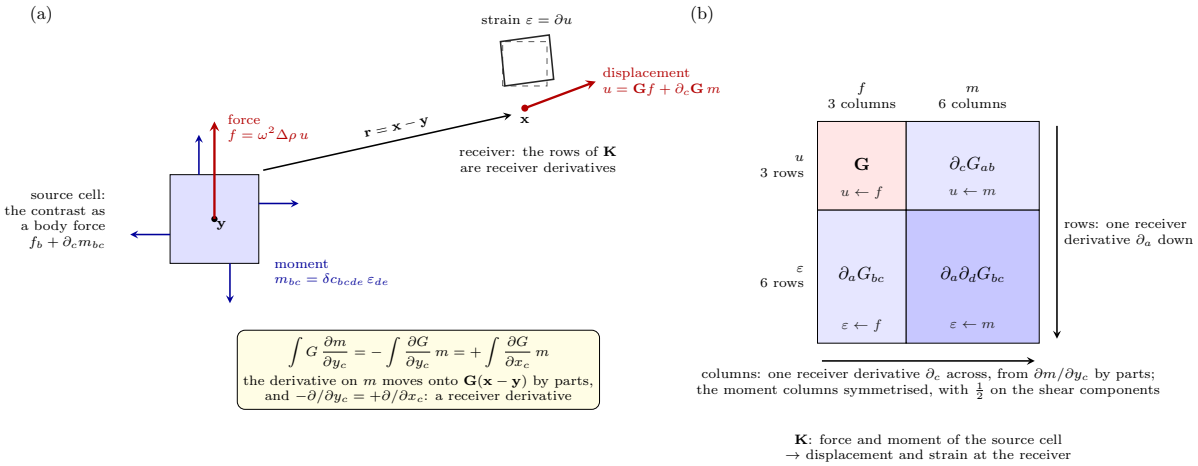


Figure 1. Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m, the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε. The moment's derivative moves onto $\mathbf{G}(\mathbf{x}-\mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked y; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled r equals x minus y runs from the cell to a red dot marked x at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y-derivative of m equals minus the integral of the y-derivative of G times m, equals plus the integral of the x-derivative of G times m. On the right, a two-by-two grid of shaded blocks with rows labelled u, three rows, and strain, six rows, and columns labelled f, three columns, and m, six columns; the blocks read G, the first derivative of G, the first derivative of G, and the second derivative of G; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

[40], the same field reads $M_{bc}\, \partial G_{ab}/\partial y_c$, the source derivative, with $M = -m$. The two are one convention written two ways; what matters is that the derivative in the kernel and the sign in the contrast operator are chosen together. The nine-component kernel follows (Fig. 1): its force columns are $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are those entries and its strain rows their receiver derivatives, symmetrised in the same way and doubled on the shear rows to give engineering strain. In that convention the contrast operator is

$$\mathbf{\Delta} = \operatorname{blockdiag}\big(\omega^2 \Delta\rho\, \mathbf{I}_3,\ \Delta\mathscr{C}\big), \tag{5}$$

where $\Delta\mathscr{C}$ is the 6×6 Voigt form of δc, the contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

Averaging, and collocation. Three operations in this paper are averages over a cell, and they are distinct. The *source-cell average* integrates the kernel over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r}-\mathbf{y})\, \mathrm{d}\mathbf{y}$ is the field at $\mathbf{r}$ of a uniform polarisation of the source cell. Every scheme in this paper uses it, and §5 shows it to be exact. *Collocation* fixes where the discrete

equation is imposed: at the receiving cell's centre, with the internal field of each cell uniform, so that the unknown is the field at the centre. The *receiver-cell average* imposes the equation in the mean over the receiving cell instead; with a uniform internal field that is the Galerkin scheme of degree zero, called the mean-only voxel below. The two are both *uniform-field* voxels, a term that refers to the internal field and not to the test, and they differ only in the test function (a delta at the centre against the cell's indicator) and in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*, finally, is an average of the material, not of the field: where the medium varies within a cell, the contrast is projected onto the cell's basis, its mean for degree $r = 0$, rather than sampled at the centre (§§8, 10). In the language of the coupled-dipole method, the point dipole is collocation with the point kernel, the integrated Green tensor adds the source-cell average, and the filtered coupled dipoles replace the point source by one of finite extent.

The laterally uniform layer. A layer of thickness D is filled with n planes of voxels, $d = D/n$, and illuminated at normal incidence. The field is then laterally uniform, only the zero-lateral-wavenumber (specular) component of the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\,\hat{\mathbf{z}}), \tag{6}$$

where the sum runs over the lattice vectors $\mathbf{R} = d\,(0, p, q)$, p, q integers, that join the centres of the voxels in one plane (a square lattice of pitch d), and m is the separation of the two planes in units of d (Fig. 2a,b). The question is whether, and how fast, this chain converges to the continuous layer as $n \to \infty$ at fixed D.

Implementations. Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. "The package" below is the Python/Fortran implementation of the voxel scheme (`cubic_scattering`): its cube T-matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.

3 The continuous layer and its thin-layer expansion

At normal incidence a plane force drives a one-dimensional problem per wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane Green's function $g(z) = \frac{\mathrm{i}}{2Mk}\mathrm{e}^{\mathrm{i}k|z|}$. The layer's reflection and transmission follow from continuity of displacement and traction at its faces; they agree with the stratified propagator used as the reference in this work to 2.5×10^{-8}, P and S (`ContinuumLimit_Reference.wl`).

For a layer of thickness D with modulus M_1 and density ρ_1 in a background M_0, ρ_0, write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their ratio and $k_1 = \omega\sqrt{\rho_1/M_1}$ for the layer's

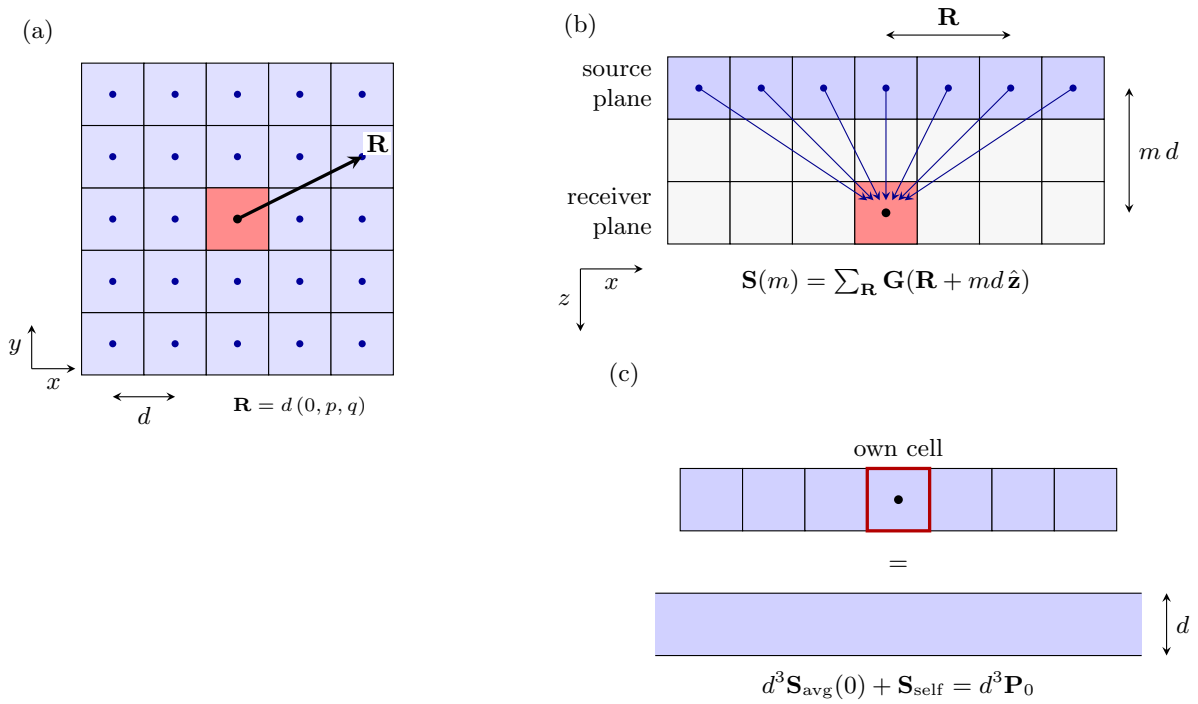


Figure 2. The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d, and a lattice vector $\mathbf{R} = d\,(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d, the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled R runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d. At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation m d marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d, a uniform plate.

wavenumber. The reflection coefficient expands as

$$R = \frac{\mathrm{i}k_1 D}{2}\left(\zeta - \zeta^{-1}\right) + \frac{(k_1 D)^2}{4}\left(\zeta^{-2} - \zeta^{2}\right) - \frac{\mathrm{i}(k_1 D)^3}{24}\left(\zeta - \zeta^{-1}\right)\left(3\zeta^{-2} + 2 + 3\zeta^{2}\right) + \frac{(k_1 D)^4}{48}\left(\zeta^{2} - \zeta^{-2}\right)\left(3\zeta^{2} - 2 + 3\zeta^{-2}\right) + O(D^5), \tag{7}$$

exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each coefficient symbolically and the truncated series against the exact layer). The first term is

$$\frac{\mathrm{i}\omega D}{2}\sqrt{\frac{M_0}{\rho_0}}\left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0\left(\frac{1}{M_1} - \frac{1}{M_0}\right)\right]: \tag{8}$$

to first order a thin layer is an interface across which the traction jumps by the excess inertia and the displacement by the excess compliance. At oblique incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A}\mathbf{b}$ for the P–SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$

as the jump $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients R_{PP}, R_{PS} to 10^{-12}; the series truncated at $O(D^k)$ converges at exactly order k; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η tends to the linear-slip interface as $D \to 0$. These expansions are the target a single plane of voxels must reproduce order by order. A plane of voxels that carry Legendre moments of their internal field to degree p reproduces Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§7): the mean-only voxel reproduces the equivalent interface and the D^2 term, the first-moment voxel all four terms shown.

4 The specular sums of the point Green's tensor

Why transform the sum. The specular sum (6) adds, at a receiver, the field of every voxel of a plane m planes away. In that form it is a poor object to study. The Green's tensor decays slowly with distance, so the sum converges slowly, and it hides the one question that matters here: which part of it is the continuum, and which part is the discretisation. The Poisson summation formula [41] answers both at once, and is the standard tool for lattice sums of wave fields [42]. For a function f on the plane and the square lattice of pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \qquad \hat{f}(\mathbf{g}) = \int f(\mathbf{x})\, \mathrm{e}^{-\mathrm{i}\mathbf{g}\cdot\mathbf{x}}\, \mathrm{d}^2x, \qquad \mathbf{g} = \tfrac{2\pi}{d}(p, q), \tag{9}$$

with p, q integers: a sum of samples of f on the lattice equals a sum of samples of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3). Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g};\, md), \tag{10}$$

with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor between planes a distance z apart. Each term has a physical meaning. Transformed laterally, the field of a plane of sources is a superposition of plane waves, one for each lateral wavenumber, and the lattice selects the wavenumbers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$, far beyond the wavenumber of the field when the voxels are small compared with the wavelength, so it is evanescent and decays away from the plane as $\mathrm{e}^{-|\mathbf{g}|md} = \mathrm{e}^{-2\pi m\sqrt{p^2+q^2}}$, independently of d. The transform therefore splits the lattice sum exactly into the continuum and a discretisation correction, and the correction converges after a few orders.

The local term that survives. In the strain-from-moment block the evanescent orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume does *not* vanish as $d \to 0$: at $m = 1$ it is eleven times the continuum term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald lattice sum to 5×10^{-13}). This surviving term is static. Summed over all planes in slab order, the static lattice sum of the point kernel is exactly the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$ on the normal shears, per unit volume) plus a cubic-symmetric remainder (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel

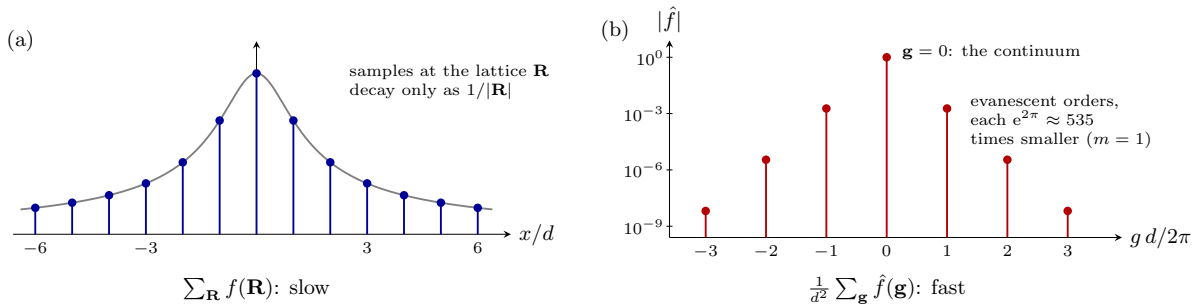


Figure 3. The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

therefore reproduces the continuum only if that remainder is cancelled by the voxel's own self term.

5 The tiling identity

Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} - \mathbf{y})\, \mathrm{d}\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel. Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \operatorname{sinc}(g_x h)\, \operatorname{sinc}(g_y h), \tag{11}$$

which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\operatorname{sinc}(p\pi) = 0$ for integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane, $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ: a partition of unity (Fig. 4). Two exact statements follow.

Between planes. For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0;\, md)\, \operatorname{sinc}(k_W h), \tag{12}$$

the continuum's plane-wave term times the vertical form factor. The package's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_AveragedSums.wl`).

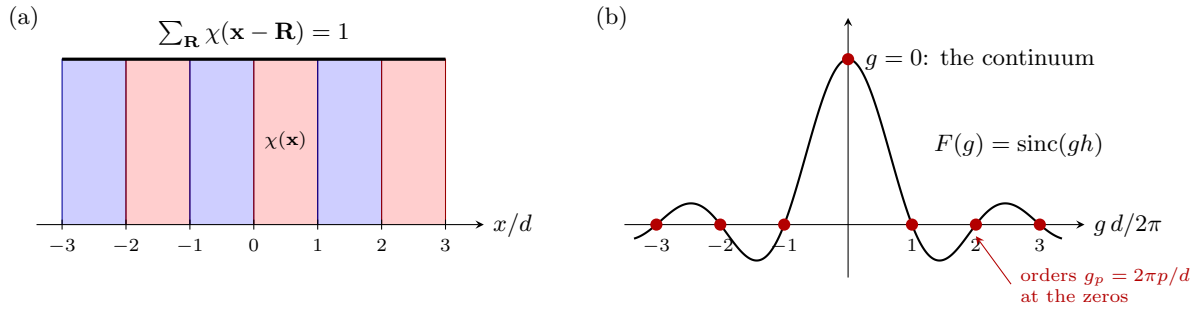


Figure 4. Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \mathrm{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

Within a plane. The sum over *all* cells of the plane, the voxel's own included, is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3\,\mathbf{S}_{\mathrm{avg}}(0) + \mathbf{S}_{\mathrm{self}} = d^3\,\mathbf{P}_0, \qquad \mathbf{S}_{\mathrm{self}} = \int_{\mathrm{cube}} \mathbf{G}(-\mathbf{y})\,\mathrm{d}\mathbf{y}, \tag{13}$$

where $\mathbf{P}_0 = d^{-3}\int_{-h}^{h} \mathbf{G}_{\mathrm{plate}}(-z')\,\mathrm{d}z'$ is elementary: in the one-dimensional transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A\,(\mathrm{e}^{\mathrm{i}kh} - 1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the three normal strain-from-moment entries, whose local weights $-1/M_P$ and $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for all 21 independent components of the strain-from-moment tensor to 4×10^{-9}, with every cell integral evaluated analytically (below).

Consequence: an exact kernel with no lattice sum. At normal incidence the same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\mathrm{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\mathrm{self}}/d^3, \tag{14}$$

with no Ewald splitting, near shell or asymptotic tail. The route it replaces (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$ Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4} at its default settings, falling to 2.6×10^{-5} only when both the shell radius and the quadrature order are raised; Eq. (14) is now the package's kernel at normal incidence.

Cell integrals without quadrature. Write the static (Kelvin) tensor as $G_{bc} = (c_A + c_B)\,\delta_{bc}/r - c_B\,\partial_b\partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$. Every cell integral of its derivatives is

a cell integral of derivatives of $1/r$ and r. One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$, which never contain the origin, and there the two-dimensional antiderivative in the face coordinates is continuous, so the face integral is a sum over the face's corners. The distributional term at $r = 0$ is carried by the divergence theorem itself. (A triple antiderivative evaluated at the cube's corners is *not* usable for repeated derivatives: its arctangent terms jump across the coordinate planes through the origin, and differentiating twice there discards exactly that term.) The self term so obtained reproduces the classical closed form of the cube's depolarisation [43, 44] to machine precision.

The collocation scheme. The simplest voxel takes its internal field as uniform. Its single-site T-matrix is then the uniform-strain closure $\mathbf{T} = d^3\,\boldsymbol{\Delta}\big(\mathbf{I} - \mathbf{S}_{\mathrm{self}}\boldsymbol{\Delta}\big)^{-1}$, with $\mathbf{S}_{\mathrm{self}}$ the nine-component self term of the cube, whose moments of the Green's tensor are in closed form (`Mathematica/CubeMomentCore.wl`); for internal state e the voxel's own field at its centre is $\mathbf{S}_{\mathrm{self}}\boldsymbol{\Delta}\,e$, and $e = \psi + \mathbf{S}_{\mathrm{self}}\boldsymbol{\Delta}e$. With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\mathrm{self}}\boldsymbol{\Delta})e$ into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\mathrm{all}}(i,j)\,\boldsymbol{\Delta}\,e_j, \qquad \mathbf{K}_{\mathrm{all}}(i,i) = \mathbf{P}_0, \quad \mathbf{K}_{\mathrm{all}}(i,j) = \mathbf{S}_{\mathrm{avg}}(i-j), \tag{15}$$

an identity checked on the matrices to 4×10^{-16}. Every block of $\mathbf{K}_{\mathrm{all}}$ is an *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)), so Eq. (15) is precisely the collocation of the continuum Lippmann–Schwinger equation with a strain piecewise constant over voxels. It does not contain $\mathbf{S}_{\mathrm{self}}$ at all.

The package's cube T-matrix is this closure statically, to 2×10^{-5} at $k_S h = 0.01$, and departs dynamically by about $0.25(kh)^2$ through its far-field form-factor corrections (`ContinuumLimit_CubeT.wl`); by Eq. (15) that departure cannot enter the closure scheme.

6 Convergence, and the closed-form error

The experiment. Every convergence test of this paper is the same experiment, varied in one respect at a time, and it is set out here once.

1. *The medium.* The background has $\alpha = 5000$ and $\beta = 3000\,\mathrm{m\,s^{-1}}$ and $\rho = 2500\,\mathrm{kg\,m^{-3}}$. The layer occupies $0 < z < D$ with $D = 2\,\mathrm{m}$, and differs from the background by $\Delta\lambda = 2\,\mathrm{GPa}$, $\Delta\mu = 1\,\mathrm{GPa}$ and $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$: 6.4% in the P modulus, 4.4% in the shear modulus and 4% in the density.

2. *The source.* A plane force at $z_s = -12\,\mathrm{m}$, above the layer, radiates a plane wave towards it. The incident field in the layer is then known in closed form, and it is smooth there because the source is outside.

3. *What is measured.* The scattered displacement, total less incident, at two observers outside the layer: at $z = -1\,\mathrm{m}$ above it, where it is the reflected wave, and at $z = 4\,\mathrm{m}$ below it, where it is the change the layer makes to the transmitted wave. Outside the layer the

scattered field is a single outgoing wave, so one observer on each side holds everything; the two are reported separately, as reflection / transmission.

4. *The exact answer.* The reflection and transmission of the layer from continuity of displacement and traction at its two faces (§3); for a layer whose properties vary with depth, from the differential equation integrated to 30 digits.

5. *The error.* $e = |u_{\text{scheme}} - u_{\text{exact}}|/|u_{\text{exact}}|$ at each observer: relative to the scattered field, not to the total, so that an error of 10^{-6} is a millionth of what the layer scatters.

6. *The refinement.* The layer is cut into n planes of voxels of side $d = D/n$, with n from 1 to 32, at $\omega = 60$, 300 and $600\,\mathrm{rad\,s^{-1}}$, for which $k_P D = 0.024$, 0.12 and 0.24 and $k_S D = 0.04$, 0.2 and 0.4. Three frequencies show that an order is not an accident of one.

7. *The arithmetic.* The errors of the higher-degree voxels fall below the round-off of double precision, so the symbolic implementation computes them to 40 digits; the independent numerical implementation, in double precision, agrees with it down to its own floor of about 10^{-13}.

The later sections change one item each: the incident wave to SV or SH and its angle (§7), the medium to a stratified or a graded one and the background to a stratified one (§8), and the body to a sphere (§10).

Second order. For a layer $D = 2\,\mathrm{m}$ with the contrast $\Delta\lambda = 2\,\mathrm{GPa}$, $\Delta\mu = 1\,\mathrm{GPa}$, $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$ in a background $\alpha = 5000$, $\beta = 3000\,\mathrm{m\,s^{-1}}$, $\rho = 2500\,\mathrm{kg\,m^{-3}}$, illuminated by a P plane force, the reflected displacement of the chain converges to the continuous layer's at order 2.000 for $n = 4 \to 16$ at $\omega = 60$, 300 and $600\,\mathrm{rad\,s^{-1}}$, with the collocation (15) and with the package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against the exact solution and n the refinement; the orders in the legend of Fig. 6 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted, with their standard errors. The two agree where a sequence is asymptotic and differ where it is not, which the apparent orders show.

The leading error in closed form. In Eq. (15) the kernel is integrated exactly; the only approximation is the internal field's variation across a voxel. At first order in the contrast the discrete scattered field is therefore a midpoint rule applied to the continuum's Born integral, with the kernel's cell integral exact. For reflection the integrand carries the two-way phase e^{2ikz}; for transmission it is flat. With k the background wavenumber of the wave concerned (k_P for an incident P wave, k_S for an incident S wave), R the reflection coefficient and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\mathrm{disc}}}{R} = \frac{\mathrm{sinc}(kh)}{\mathrm{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \qquad \frac{T^{\mathrm{sc}}_{\mathrm{disc}}}{T^{\mathrm{sc}}} = \mathrm{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \tag{16}$$

confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms and those of Eq. (19) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §7). At the full contrast the reflection constant is unchanged (0.124 against 1/8); the transmission constant grows to −0.092.

Why a uniform-field single site stops at second order. The two constants have opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can cancel both. The package *T*-matrix illustrates it: its form-factor corrections reduce the reflection error by a factor 4.45 and *increase* the transmission error by 1.7, at every n. The missing term in Eq. (16) is the first moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

7 Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel j, with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) = P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, …), the field is $w = \sum_b c_{b,j}\,\phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the polarisation is $q = (\omega^2 \Delta\rho\, u,\ \Delta M\, \varepsilon)$ with $\Delta M = \Delta\lambda + 2\Delta\mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z - z')\, q(z')\, \mathrm{d}z', \qquad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \tag{17}$$

with the same functions on every voxel gives

$$\Big(\int_i \phi_a^2\, \mathrm{d}z\Big) c_{a,i} = \int_i \phi_a w^0\, \mathrm{d}z + \sum_{j,b} \Big[\iint_{i,j} \phi_a(z)\, \mathbf{K}(z - z')\, \phi_b(z')\, \mathrm{d}z\, \mathrm{d}z'\Big] \mathbf{\Delta}_1 c_{b,j}, \tag{18}$$

$\mathbf{\Delta}_1 = \mathrm{diag}(\omega^2\Delta\rho, \Delta M)$. Between voxels the double integral is separable, a product of Legendre moments of $\mathrm{e}^{\pm \mathrm{i}kz}$; within a voxel it is split at $z' = z$ and the local term contributes $-\int_i \phi_a \phi_b\, \mathrm{d}z / M_P$. Every entry is closed-form. Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately

Table 1. Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300\,\mathrm{rad\,s^{-1}}$, against the number of voxel planes n: collocation, the mean-only voxel and the first-moment voxel of Eq. (18). The order is 4.00 at $\omega = 60$ and $600\,\mathrm{rad\,s^{-1}}$ too.

n	collocation	mean only	mean and first moment
1	1.8×10^{-3} / 1.3×10^{-3}	1.2×10^{-3} / 1.8×10^{-3}	2.8×10^{-7} / 2.8×10^{-7}
4	1.1×10^{-4} / 8.3×10^{-5}	7.5×10^{-5} / 1.1×10^{-4}	1.1×10^{-9} / 1.1×10^{-9}
16	7.0×10^{-6} / 5.2×10^{-6}	4.7×10^{-6} / 6.9×10^{-6}	4.2×10^{-12} / 4.2×10^{-12}
order	2 / 2	2 / 2	4 / 4

than a single plane of uniform ones. The collocation column is the scheme of Eq. (15) and reproduces the chain of the full 9×9 system to all printed digits.

Higher moments. Carrying the Legendre moments to degree p changes nothing in the construction: the moments and the same-cell double integrals of $P_b(s/h)$ against exponentials are again closed-form, and for $p \le 1$ the general-degree solver reproduces the chain above exactly. The second-moment voxel ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment.wl`); the second-moment voxel's order is 6.00 at $\omega = 60$, 300 and 600 $\mathrm{rad\,s^{-1}}$. The errors fall to 10^{-22}, so they are computed in 40-digit arithmetic.

Table 2. Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300\,\mathrm{rad\,s^{-1}}$, for voxels carrying Legendre moments to degree p. The order is the mean over the halvings $n = 2 \to 4 \to 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	2.8×10^{-7} / 2.8×10^{-7}	2.8×10^{-11} / 4.2×10^{-11}	1.5×10^{-15} / 1.5×10^{-15}
2	1.7×10^{-8} / 1.7×10^{-8}	4.4×10^{-13} / 6.5×10^{-13}	6.0×10^{-18} / 6.0×10^{-18}
4	1.1×10^{-9} / 1.1×10^{-9}	6.9×10^{-15} / 1.0×10^{-14}	2.4×10^{-20} / 2.4×10^{-20}
8	6.7×10^{-11} / 6.7×10^{-11}	1.1×10^{-16} / 1.6×10^{-16}	9.2×10^{-23} / 9.2×10^{-23}
order	4 / 4	6 / 6	8 / 8

The leading error in closed form. Without scattering, the Galerkin solution is the L^2 projection Π of the incident field onto each voxel's basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs}, $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^{h} e^{iqs}\,\Pi[e^{iks}]\,\mathrm{d}s}{\int_{-h}^{h} e^{i(q+k)s}\,\mathrm{d}s}, \tag{19}$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (18) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6}. The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1)\, j_j(kh)\, P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O\big((kh)^{p+1}\big)$; its moment against e^{iqs} is $2h\, i^{p+1} j_{p+1}(qh) = O\big((qh)^{p+1}\big)$. With $j_n(x) \approx x^n/(2n+1)!!$ and $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1}\big[(2p+3)!!\big]^2} = \frac{\big[(p+1)!\big]^2}{(2p+2)!\,(2p+3)!}, \qquad c_0 = \tfrac{1}{12},\ c_1 = \tfrac{1}{720},\ c_2 = \tfrac{1}{100800}, \tag{20}$$

Table 3. The leading error of the discrete layer in closed form, Eq. (19): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n.

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and transmission for odd p, opposite signs for even p. The first-moment voxel's leading error therefore does not depend on the direction in which the field is observed. Eq. (20) is checked symbolically for $p = 0$ to 5 (every lower power of kd vanishes) in Mathematica and, independently, in sympy.

One plane of voxels and the thin-layer series. Refining at fixed thickness is one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$, $d = D$) the series in D of the reflection coefficient, computed exactly at the full contrast, coincides with that of the exact layer, Eq. (7), term by term through D^{2p+2} and first differs at D^{2p+3}, for $p = 0$ to 3 and for a layer both stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.wl`). Each further Legendre moment adds two terms of the thin-layer series.

In three dimensions. The three-dimensional first-moment voxel carries the nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (18) (`ContinuumLimit_Reduction3D.wl`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and x/h, y/h are odd there. The mean and the z-moment carry weights that are constant laterally, so their lateral form factor $\operatorname{sinc}(g_x h)\operatorname{sinc}(g_y h)$ vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = \mathrm{i}\,(gh\cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = \mathrm{i}(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift, $\operatorname{sinc}\big((2\pi p/d + k_x)h\big) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds only to $O(k_\parallel h)$.

Other angles, other waves, and degree two. The orders and their constants hold beyond normal incidence and incident P. At oblique incidence, where the tiling identity holds only to $O(k_\parallel h)$ and the gradient channels couple through the evanescent orders, the mean-only and first-moment voxels remain second and fourth order in the reflected P and the converted S wave; for incident SV and SH waves the orders, and with $k = k_S$ the constants, are those of incident P; and the voxel of degree two converges at sixth order for P, SV and SH waves, its error on four

planes at normal incidence being the $c_2(kd)^6$ of Eq. (20) to 4%. Appendix B gives the spectral construction, Eq. (40), the tests, and the second implementations that check them; Figure 6(a,b) shows the convergence.

8 A heterogeneous layer

A heterogeneous medium is not a layer of identical cells. The simplest heterogeneous case with an exact answer is a stratified model built from layers of constant properties, with real interfaces between them: the form of many layered crustal velocity models. The last test takes such a model. A layer $D = 4\,\mathrm{m}$ thick is divided into eight model layers of $0.5\,\mathrm{m}$, and in each one $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within ±15% of the background (the values are listed in `ContinuumLimit_Heterogeneous.wl`). The result is a random stratification with physical contrasts. The voxels refine the model's layering, with m planes of voxels in each model layer, so no voxel straddles a model interface: the natural discretisation when the model's interfaces are known. At $\omega = 1500\,\mathrm{rad\,s^{-1}}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field reverberates between the model interfaces. The schemes are those of §§6–7, with the contrast now changing from plane to plane; nothing else in them changes. In the terms of the experiment of §6, the test is this.

1. *The medium.* The eight sets of contrasts are drawn once, from a fixed seed, and the same eight are used in every run, so that every grid and every scheme sees the same model.

2. *Normal incidence.* The source is the plane force at $z_s = -12\,\mathrm{m}$, and the scattered displacement is taken at $z = -1\,\mathrm{m}$, where it is the reflected wave, and at $z = D + 2 = 6\,\mathrm{m}$, where it is the change in the transmitted wave. The P problem has the modulus contrast $\Delta\lambda + 2\Delta\mu$ and the wave speed α; the S problem is the same one-dimensional problem with $\Delta\mu$ and β.

3. *Oblique incidence.* A P plane wave is incident at 20° to the vertical. What is measured is the pair of displacement amplitude vectors of the reflected P and the reflected S plane waves, R_{PP} and R_{PS}, in the specular order. The lateral sums of the coupling (§7) run over the reciprocal-lattice orders $|p|, |q| \le 8$.

4. *The exact answer.* The state of displacement and traction, two components at normal incidence and four for P–SV at 20°, is carried across each model layer by the matrix exponential of that layer's system matrix times its thickness, the eight matrices are multiplied in order, and the radiation conditions of the two half-spaces are imposed on the product. The arithmetic carries 30 to 40 digits. The reference shares nothing with the voxel scheme but the model. It is checked against a third algorithm, the reflection–transmission recursion of Kennett [45] over the nine interfaces: the same-type coefficients, R_{PP} and R_{SS} at normal incidence and R_{PP} at 20°, agree to 10^{-14} (`scripts/gate_heterogeneous_reference_vs_kennett.py`). Only same-type coefficients are compared, because they do not depend on how the up- and downgoing waves are normalised; a converted coefficient does. One sign differs, by convention: at normal incidence the one-dimensional model's

R_{PP} is a ratio of vertical displacements, the recursion's a ratio of amplitudes along each wave's own direction, and the reflected P wave travels upwards.

5. *The error.* As in §6, the difference from the exact scattered quantity relative to its size: for the amplitude vectors at 20°, $|\mathbf{R}_{\text{scheme}} - \mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$.

6. *The refinement.* Each model layer is cut into m planes of voxels, so that the layer holds $n = 8m$: $m = 1, 2, 4, 8, 16$ at normal incidence and $m = 1, 2, 4$ at 20°, where the lateral sums make each run dearer. The orders are apparent orders, from $m = 4$ to 16 and from $m = 2$ to 4.

The errors are in Table 4 and Fig. 6(c).

Table 4. Relative error against the exact stratified layer, for m planes of voxels per model layer (§8). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20°, R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20°, R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

The interfaces cost nothing. Every order is the one measured for the uniform layer. At a model interface the strain of the true field jumps. A voxel basis that is discontinuous from voxel to voxel represents that jump exactly when the interface lies on a voxel boundary, so the interface adds no error of its own. The only requirement is that voxel boundaries coincide with the model's interfaces. A voxel that straddled an interface would carry a contrast that is not uniform within it, a case not examined here.

What fourth order buys. For a reflection error of 10^{-6} at normal incidence (P), the mean-only voxel needs 347 planes of voxels, about 43 per model layer. That figure is not a run: it is the second-order law carried from the finest grid computed, 128 planes, whose error is 7.3×10^{-6}, to the target, $n = 128\,(7.3 \times 10^{-6}/10^{-6})^{1/2}$. The first-moment voxel needs 8, one per model layer, and that figure is a run: the model's own layering already gives 8.5×10^{-7}. Counting unknowns (9 per voxel against 36), that is 3123 against 288 per vertical column of voxels, a factor of eleven. In three dimensions, with the lateral grid refined as the vertical one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of unknowns by about 2×10^4. The same model has been solved independently in Python, with a contrast for each plane: the raw reflected displacements agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and at 20°, with and without the first moments. Lateral heterogeneity needs the full three-dimensional solver; it is the first extension discussed below.

A medium that varies within a cell. The stratified model is exact on its own grid: every cell is homogeneous. A smoothly varying medium is not, and sampling it as cells of constant properties is an approximation of its own. Let the contrast of cell j be its Legendre projection to degree r, $\mathbf{\Delta}(z) = \mathbf{\Delta}_0 \sum_{c\le r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §7. The product of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} = (2e+1)\left(\begin{smallmatrix} c & b & e \\ 0 & 0 & 0 \end{smallmatrix}\right)^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (18) becomes

$$\int_i\!\int_j \phi_a(z)\,\mathbf{K}(z-z')\,\mathbf{\Delta}(z')\,\phi_b(z')\,\mathrm{d}z\,\mathrm{d}z' = \sum_{c\le r}\sum_{e\le p+r} f_c^{(j)} A_{cbe}\Big[\int_i\!\int_j \phi_a(z)\,\mathbf{K}(z-z')\,\phi_e(z')\,\mathrm{d}z\,\mathrm{d}z'\Big]\mathbf{\Delta}_0 \,: \quad (21)$$

the same kernel moments, now to degree $p+r$, and nothing else new. Against the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2\rho(z)u = 0$, the order of convergence is

$$\min(2p+2,\ 2r+2), \quad (22)$$

or $2p+2$ when degree r represents the profile exactly (Table 5; `ContinuumLimit_GradedContrast.wl`). A contrast that is constant in each cell therefore caps every voxel at second order, however rich its field basis, and for the smooth profile the voxel of degree 2 with a linear contrast has the error of the voxel of degree 1 to within 0.06%. At first order in the contrast this is what one expects: the contrast's projection error is orthogonal to the polynomials of degree r, so against the smooth product of the incident and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$ the contrast must be represented to the degree of the field. The closed forms of the construction are tested

Table 5. A contrast that varies within the cells: the field of each voxel to Legendre degree p, its contrast to degree r. Relative reflection error at $n = 16$ planes and the order ($n = 4 \to 8 \to 16$), $\omega = 300\,\mathrm{rad\,s^{-1}}$, for the profiles $1+(2z/D-1)/2$ (linear) and $1+\frac{1}{2}\sin(2\pi z/D)$ (smooth) times the contrast of §6. Predicted: Eq. (22), and $2p+2$ for the linear profile once $r \ge 1$.

		linear profile		smooth profile	
p	r	error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

one by one by an independent implementation (`scripts/crosscheck_graded_contrast.py`): the Legendre moments of the kernel against $2h\,\mathrm{i}^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the linearisation coefficients against the $3j$ symbols (exactly), and the same-cell double integrals against 30-digit quadrature (1.4×10^{-16}). It then assembles the scheme without linearisation, evaluating the projected contrast at the quadrature nodes, against its own 30-digit solution of the graded layer; its errors agree with those of Table 5 to 10^{-3}, and at $\omega = 1200\,\mathrm{rad\,s^{-1}}$ it measures the orders of Eq. (22) again.

Which basis sets the error. Equation (22) gives the order. The size of the error, and which of the two bases it belongs to, appear when the scattered field is separated by its order in the

contrast: the Born term T_1, and the second-order term T_2 in which the field scattered once is scattered again.

Appendix F gives both terms for the exact layer in closed form (Eqs. (47)–(52)) and for the scheme, where the first needs no solve (Eq. (53)), and shows how both are read from solutions at four scaled contrasts (Eq. (54)). The terms must be separated: at the full contrast the terms of third and higher order are 7% of the second and carry errors of their own.

With $f(z)$ the profile and Π_q the L^2 projection onto the Legendre polynomials of degree at most q in each cell, define the *relative projection error* of the medium at degree q,

$$\mathscr{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 \, \mathrm{d}z}{\int f^2 \, \mathrm{d}z} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \qquad \|f\|^2 = \int f^2 \, \mathrm{d}z : \tag{23}$$

the fraction of the squared norm of the profile that polynomials of degree q on the cells cannot represent. It is a property of the medium and the grid alone; it involves neither the wave nor the solution. For the smooth profile and every one of the nine pairs with $p, r \le 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathscr{E}_{\min(p,r)} \tag{24}$$

to within 1.3% at $\omega = 300\,\mathrm{rad\,s^{-1}}$ on 4 to 16 cells: the ratio of the two sides is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2. The error of the Born term is smaller by two to six orders of magnitude. The error of the whole is therefore that of T_2, the projection error times $|T_2|/|T_1|$ (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7}, and Table 5 has 1.4×10^{-7}.

Equation (24) is exact in the long-wave limit (Appendix F). There the strain scattered once is the profile itself, a cell holds it as $\Pi_{\min(p,r)} f$, and the second scattering takes place in the cell of the first: the law is a statement about the single site, whose T-matrix is exact for the cell as represented and whose error is that the cell is not the medium. Measured, the part of T_2 between different cells carries at most 0.002 of the error, and the departure from the law falls as $(kD)^2$.

Three consequences follow. The two bases must be matched: a contrast of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$ have the errors of $(0, 0)$ to three digits), and a field of higher degree than the contrast improves only the Born term. The wavefield basis is there to carry what the medium imprints on the field, not only the wave: at these frequencies the wave itself varies little across a cell. And the error is known from the medium, before anything is solved: $\mathscr{E}_q$ is a property of the profile and the cells. It vanishes when the medium is piecewise constant on the cells, which is the stratified model above; there the contrast needs no more than a constant in each cell, and the degree of the field alone sets the order.

A stratified background. Everything above takes the background to be a whole space. The method is meant for a background that is plane stratified, with the voxels carrying the departure from it, and the results must hold there. They do, in the one case tested: normal incidence, P, with the background two half-spaces joined at $z = z_I$ below the layer, the lower one faster and denser (displacement reflection coefficient −0.19). The voxels then interact through the Green's

function of that background,

$$g(z,z') = \frac{\mathrm{i}}{2M_P k}\Big[\mathrm{e}^{\mathrm{i}k|z-z'|} + \rho_I\,\mathrm{e}^{\mathrm{i}k(2z_I-z-z')}\Big], \qquad z,z' < z_I, \tag{25}$$

with ρ_I the reflection coefficient of the interface: the whole-space term and a reverberation that is smooth and carries no local term. The incident field is the field of the source in the background, and the exact answer is the reflection of the whole stack less that of the background. With the interface 3 m below the layer, and again with the lowest voxel against it, the scheme converges at order 2.00, 4.00 and 6.0 for $p = 0$, 1 and 2 on the uniform layer, and at 1.98, 3.98 and 5.98 on the smooth profile with $r = p$. The first-order term is predicted to round-off (5×10^{-16}): it is the sum over four paths, the incident wave travelling down or up and the scattered wave leaving up or down, of the exact first-order term of each path times the factor $\sum_{a\le p}(2a+1)\,j_a(k_{\mathrm{in}}h)\,j_a(k_{\mathrm{out}}h)/j_0\big((k_{\mathrm{in}}-k_{\mathrm{out}})h\big)$ for that path's pair of wavenumbers, whose leading terms are the constants of Eq. (20). And the relative error of T_2 over $\mathscr{E}_p$ is 1.00 to 1.02 at $kD = 0.12$ on 4 to 16 cells for all three degrees, departing from one by 1.1×10^{-4} and 3×10^{-5} at $kD = 0.012$ for the two positions of the interface: Eq. (24) holds as in the whole space (`scripts/measure_layer_stratified_background.py`; the exact stack is computed in 30-digit arithmetic). The reason is the one given for Eq. (55): the reverberation adds no local term, so the second scattering in the cell of the first is unchanged. This check is by one implementation against an exact solution.

Oblique incidence, and a three-layer background. The same holds at oblique incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer background in which the layer of voxels reverberates between two interfaces (Appendix C). At 20° the orders are 2, 4 and 6 for $p = 0$, 1 and 2 in all thirty-six cases (Table 13), and the errors are close to those of the whole space: the background changes the field the voxels must represent, not how well they represent it. Both terms of the error remain known in advance. The scheme's first-order term equals a closed-form prediction, built from the thirty-six pairs of incident and radiated plane waves (Eq. (43)), to 2×10^{-15}; and the law of Eq. (24) holds to 3% for $p \le 1$ and 6% for $p = 2$ at $k_S D = 0.2$, and to 0.6% at $k_S D = 0.067$, in every background and for every wave type (Table 14).

Part II

Three dimensions

9 Galerkin cells in three dimensions

Which cells. The cells of Part II are the Legendre voxels of §7 on a cubic grid (Appendix E), whose single site is that of Nestor [5]. They are Galerkin: each holds the L^2 projection of the field and of the contrast onto its own polynomials. Two of the results below rest on that. The term of first order needs no solve because a Galerkin cell holds the projection of the incident

wave (§11.1); and the second term is governed by the projection error of the medium because the cells hold its L^2 projection (§§11.3–11.6).

Cells. The body is covered by cubes: cell ℓ has centre $\mathbf{x}_\ell$ and half-width h_ℓ (side $d_\ell = 2h_\ell$). The tests below use uniform grids, $h_\ell = h$; nothing in the error law requires it, and its statements hold cell by cell on a grid of unequal cubes. With $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_\ell)/h_\ell$ the local coordinate, the cell carries the nine-component state (displacement and strain) in the Legendre products $Q_a(\boldsymbol{\xi}) = P_{a_1}(\xi_1)P_{a_2}(\xi_2)P_{a_3}(\xi_3)$ of total degree $|a| = a_1 + a_2 + a_3 \le p_\ell$, which is 1, 4 or 10 functions for $p_\ell = 0$, 1 or 2, and the contrast in the same products to degree r_ℓ. A cell with $p = r = 0$ is a constant cell: a constant medium and a constant field. A cell with $p = r = 1$ is a degree-one cell (the constant and the three linear polynomials, ξ_1, ξ_2, ξ_3), and one with $p = r = 2$ a degree-two cell (those four and the six quadratic products, $P_2(\xi_i)$ and $\xi_i\xi_j$ with $i < j$; ten in all).

The system. Tested with the same functions, the volume integral equation becomes

$$M_\ell\,\psi_\ell - \sum_m \mathbf{K}(\ell, m)\,E_m\,\psi_m = \langle Q, \psi^0\rangle_\ell, \tag{26}$$

with ψ_ℓ the coefficients of the state in cell ℓ, M_ℓ the Gram matrix of its functions, E_m the projected contrast of cell m acting on its field, ψ^0 the incident wave and $\mathbf{K}(\ell, m)$ the block that couples the two cells: the Green's tensor integrated against the functions of both [46].

10 A sphere: what the layer predicts in three dimensions

The layer and the stratified layer are laterally uniform. A sphere is not, and elastic scattering by a sphere, homogeneous [47, 48] or radially varying [49], has an exact solution: it is the usual benchmark for volume-integral and discrete-dipole methods. This section solves two spheres with the full three-dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy–Lax system is solved with the cell-averaged coupling by an FFT-accelerated iterative solver, and a P wave is incident. For the graded sphere the voxel is the mean-only voxel of §7 in three dimensions: each cell carries the mean of its nine-component state, tested with the same constant, and the L^2 projection of its contrast, which is its cell mean; every cell that overlaps the sphere is kept, and the cells are coupled by cell-to-cell double integrals of the Green's tensor, one block per orbit of the cube group (Appendix E). For the sharp sphere the voxel is the package's cube T-matrix, which differs from the collocation closure only through its far-field corrections (§5) and has the same order, each cell keeping the contrast of its centre. The far field is computed from the solution for the incident plane wave itself, each voxel's displacement and strain carrying the plane-wave phase at its centre; adding to that phase a linear variation about the scatterer's centre, as a Taylor expansion of the incident field would, counts the wave's variation twice and leaves an error of order $(ka)^2$ that no refinement removes. In the terms of the experiment of §6, the test is this.

1. *The body.* A sphere of radius $a = 10\,\mathrm{m}$ (Fig. 5) in the background of §6, carrying the contrast

of that section, at $\omega = 150$ and $300\,\mathrm{rad\,s^{-1}}$, for which $k_S a = 0.5$ and 1 and $k_P a = 0.3$ and 0.6.

2. *The incident wave.* A P plane wave travelling along $+z$.

3. *The grid.* The cube of side $2a$ that circumscribes the sphere is cut into n_{sub}^3 cells of side $2a/n_{\mathrm{sub}}$; which of them are kept is stated with each voxel above.

4. *The solve.* The iteration is carried to a relative residual of 10^{-11}, far below every discretisation error reported, so that the errors are the scheme's and not the solver's.

5. *The exact answer.* The partial-wave series of the sphere, summed to order 10 at $k_S a = 0.5$ and 11 at $k_S a = 1$, and evaluated as the exact scattered displacement at the observers, not as its asymptotic form.

6. *What is measured.* The scattered displacement vector at nine scattering angles, equally spaced from 11° to 169° in the z–x plane: forward and backward scattering and everything between, P and S together.

7. *The error.* The largest component of the difference between the scheme's far field and the exact displacement over the nine angles, divided by the largest component of the exact one. It is a maximum, not an average, so that no angle is hidden.

8. *The distance.* The solution of the scheme is the field inside the voxels, and the scattered field outside follows from it at any distance. For the far-field test it is read with the asymptotic formula, which keeps the term in $1/r$ alone: a choice of this test, not a property of the scheme. The reference is the exact field, so the two differ by the near-field terms, of relative size $1/(kr)$, however good the scheme. The graded sphere is therefore observed at 5×10^8 radii, where those terms are 10^{-8} of the peak: at 5×10^4 radii they are 1.4×10^{-4}, which is 4% of the discretisation error at 12 voxels across and most of it at 24, and would be read as a failure to converge. The sharp sphere, whose errors are a hundred times larger, is observed at 5×10^4 radii. The near field is tested separately, with a readout that drops nothing (Table 15).

A sphere without a staircase. A voxelised sphere with a sharp surface changes its shape with every grid. A sphere whose contrast falls smoothly to zero at its surface does not: the voxels that cross the surface carry a vanishing contrast. The body here has a homogeneous core $r < a/2$ carrying the contrast of §6, and across the shell $a/2 < r < a$ the contrast falls to zero as the smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from the radial equations of Takeuchi and Saito [49], integrated across the shell from the core's regular solutions and matched to outgoing waves at $r = a$. The equations are derived symbolically from the equations of motion, and the solution is checked against a sphere of homogeneous concentric shells in closed form, by unitarity and reciprocity, and by an independent implementation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_GradedSphere.wl`, `scripts/crosscheck_`

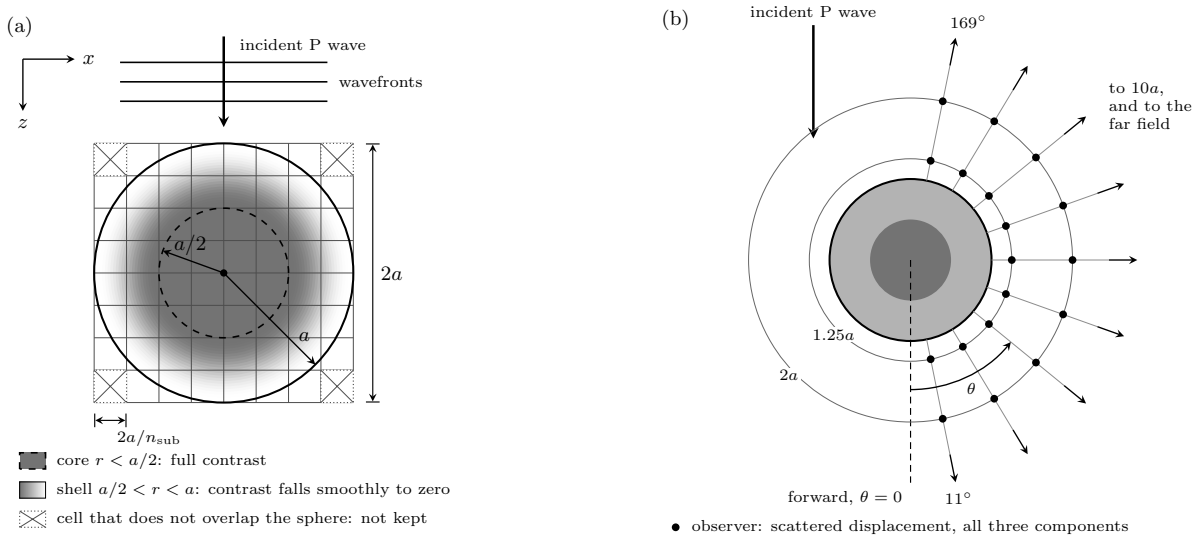


Figure 5. The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core $r < a/2$ and falls smoothly to zero across the shell $a/2 < r < a$. The circumscribing cube of side $2a$ is cut into n_{sub}^3 cells, here $n_{\mathrm{sub}} = 8$; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along $+z$. (b) The observers. The scattered displacement is taken at nine scattering angles θ, measured from the forward direction and equally spaced from 11° to 169°, on spheres of several radii about the centre, and compared with the exact displacement at the same points.

Alt text: Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

`graded_sphere.py`). Each voxel carries the mean of the contrast over its cell. Table 16 shows the error falling at second order from $n_{\mathrm{sub}} = 6$ to 24 at both frequencies (`scripts/pilot_graded_voxel_sphere.py`): the apparent order between successive grids is 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders of Fig. 6(d) are 2.045 ± 0.004 and 1.994 ± 0.004. With 24 voxels across the error is 1.8×10^{-4} and 8.3×10^{-4}. At $n_{\mathrm{sub}} = 4$ a voxel is as wide as the graded shell and the order is not yet asymptotic. This is the order the layer predicts for the mean-only voxel, and the first test of that prediction on a body with genuine three-dimensional structure.

What makes the sequence this clean is that the voxel carries a single error. The collocation voxel of the sharp sphere applied to this body, each cell at the profile's value at its centre and only the cells whose centres lie inside the sphere kept, carries two: its own internal-field error and the error of sampling the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\mathrm{sub}} = 4$ to 4.7×10^{-3}, 2.6×10^{-3}, 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6, 2.0, 3.5 and 3.6 at $k_S a = 0.5$, and at $k_S a = 1$ the apparent orders are 1.7, 2.5 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's, but its apparent order is not that of any single term and cannot be read from them. The same collocation voxel with the cell mean of the contrast, and every overlapping cell kept, gives 3.4×10^{-3}, 1.9×10^{-3} and 8.2×10^{-4} at 6, 8 and

12, apparent orders 2.03 and 2.05: the sampling of the profile was the second term, and with it removed the collocation and mean-only voxels differ only by their constants, as in Table 3. The cell mean, with every overlapping cell kept, represents the same body on every grid, and the order the layer predicts is then the one observed.

The first-moment voxel of §7, extended to three dimensions with a contrast linear in each cell, converges at fourth order on this same sphere (Appendix E): its apparent orders are 4.01 and 4.13 where the discretisation error is isolated at weak contrast, and at the full contrast its error with 16 voxels across is 4.9×10^{-6} and 5.9×10^{-6}, 82 and 317 times below the mean-only voxel's on the same grid.

The near field, an impedance march, and a sharp sphere. Three further tests are in Appendix D. Read at finite distance, down to a quarter of a radius from the surface, the graded sphere converges at the orders of the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-moment voxel, while the asymptotic readout would be in error there by 48% to 85% (Table 15). Against the impedance march of Haines et al. [50], which needs far fewer unknowns, the first-moment voxel reaches the march's accuracy with six voxels across and continues at close to fourth order below it, where the march is limited by the scattering between the spheres of the periodic array it must solve (Table 17). And a sphere with a sharp surface does not converge: its error tracks the volume of its staircase, which changes with the grid (Table 18); the smooth sphere, like the layer, measures the scheme.

11 The error in two terms

Write the scattered far field as a series in the contrast: $u = T_1 + T_2 + \dots$, with T_1 the term of first order (the Born term) and the rest the nonlinear part of the response. The error of the scheme separates the same way.

11.1 The first-order error needs no solve

At first order in the contrast Eq. (26) is $M_\ell \psi_\ell = \langle Q, \psi^0 \rangle_\ell$: each cell holds the L^2 projection of the incident wave onto its field functions. The scheme's Born term is therefore known from one pass over the cells, with no system to solve, and its difference from the Born term of a finer representation (each cell's eight children with degree-two cells) is the first-order error of the grid, cell by cell. On a grid of cells of two sizes this projected Born term equals the weak-contrast limit of the full solver to 10^{-5} of its size.

11.2 The wave factor of a cell

For a cell whose contrast is uniform the first-order error is a closed form. The cell represents the incident wave $\mathrm{e}^{\mathrm{i}\mathbf{k}_{\mathrm{in}}\cdot\mathbf{x}}$ by its projection, and it radiates towards the observer through the moments of $\mathrm{e}^{-\mathrm{i}\mathbf{k}_{\mathrm{out}}\cdot\mathbf{x}}$, with $\mathbf{k}_{\mathrm{out}}$ the P or the S wavevector in the direction of observation. Since $\int_{-1}^{1} P_n(t)\, \mathrm{e}^{\mathrm{i}qt}\, \mathrm{d}t = 2\,\mathrm{i}^n j_n(q)$, with j_n the spherical Bessel function, the cell's Born far field is its

exact Born far field times

$$1 + E_p = \frac{\displaystyle\sum_{|a|\le p} \prod_{i=1}^{3} (2a_i + 1)\, j_{a_i}(k_{\mathrm{in},i} h)\, j_{a_i}(k_{\mathrm{out},i} h)}{\displaystyle\prod_{i=1}^{3} j_0\big((k_{\mathrm{in},i} - k_{\mathrm{out},i}) h\big)}. \tag{27}$$

The sum over all degrees is the denominator (the addition theorem), so $E_p \to 0$ as p grows. Its leading term is

$$E_p = -h^{2p+2} \sum_{|a|=p+1} \prod_{i=1}^{3} \frac{(2a_i + 1)\,(k_{\mathrm{in},i} k_{\mathrm{out},i})^{a_i}}{\big[(2a_i + 1)!!\big]^2} + O(h^{2p+4}), \qquad E_0 = -\frac{(\mathbf{k}_{\mathrm{in}} \cdot \mathbf{k}_{\mathrm{out}})\, d^2}{12} + \dots \tag{28}$$

For a constant field the error of a cell therefore depends on the angle of scattering, vanishes at right angles to the incident wave, and differs between the P and the S far field. Along one axis, with $\mathbf{k}_{\mathrm{out}} = \pm\mathbf{k}_{\mathrm{in}}$ of magnitude k, Eq. (28) is $-(\pm 1)^{p+1} c_p (kd)^{2p+2}$ with $c_p = (2p + 3)/\big(4^{p+1}[(2p + 3)!!]^2\big)$: the constants $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ of the layer, Eq. (20). The layer's closed-form error is the one-dimensional case of the cell's.

Checks. Equation (27) is derived symbolically, built a second way without Bessel functions (project, multiply, integrate over the cube) and found identical for $p = 0, 1, 2$; Eq. (28) and its one-dimensional reduction are verified symbolically; and the implementation used below agrees with 30-digit values of the factor to 10^{-13} in 36 cases. Against the solver, on a homogeneous cube (which every grid of cubes represents exactly) at $k_S A = 1$ for half-side A: the Born far field divided by $1 + E_p$ agrees across $p = 0, 1, 2$ and two grids to 9×10^{-7}, where the uncorrected fields differ by up to 14%; and the Born far field of 15 cells of mixed sizes and degrees equals the sum over the cells of exact contribution times factor to 2×10^{-9}.

11.3 The rest is the projection error of the medium

Let $\mathscr{E}_r$ be the relative projection error of the medium on the grid,

$$\mathscr{E}_r = \frac{\sum_\ell \int_\ell (f - \Pi_r f)^2 \,\mathrm{d}V}{\int f^2 \,\mathrm{d}V}, \tag{29}$$

with $f(\mathbf{x})$ the profile of the contrast and Π_r the L^2 projection onto the polynomials of degree at most r on each cell. Subtract from the error of the far field the first-order error of §11.1. What is left, divided by $\mathscr{E}_r$, is one constant (Table 6). The test body is a sphere of radius a with a uniform core of radius $0.75a$ and a shell across which the contrast falls smoothly to zero, at $k_S a = 0.5$, whose exact solution is known; the error is the largest difference of the far field from the exact one over nine scattering angles, relative to its peak. For constant cells the ratio of error to projection error varies by a factor of 1.86 across the grids; the rest over the projection error varies by 1.06. For degree-one cells the first-order error is small and both ratios are near 0.0146. The constant is close to the nonlinear fraction of the exact response, $\nu = |u - T_1|/|u| = 0.0151$.

Table 6. The error of the far field in two terms, on the sphere with a graded shell. The rest is the error less the first-order error, which is computed without a solve. $\mathscr{E}$ is the relative projection error of Eq. (29) at the degree of the cells. The nonlinear fraction of the exact response is 0.0151.

cells	number	error	error / $\mathscr{E}$	rest / $\mathscr{E}$
constant ($p = r = 0$)	64	8.17×10^{-3}	0.0309	0.0149
	184	3.18×10^{-3}	0.0235	0.0151
	408	1.61×10^{-3}	0.0196	0.0154
	1256	6.77×10^{-4}	0.0166	0.0158
degree one ($p = r = 1$)	64	8.99×10^{-4}	0.0147	0.0148
	184	4.62×10^{-4}	0.0146	0.0146
	408	2.20×10^{-4}	0.0146	0.0146

This is the law of the layer (§8) carried to three dimensions: the error of the nonlinear part is that part times the relative projection error of the medium, times a factor of order one that §11.4 derives. It holds for constant cells as it does for degree-one cells, once the first-order error is accounted for; without that, constant cells appear not to follow it.

11.4 The factor of the second term in three dimensions

Why a factor appears. In the layer the relative error of the term of second order in the contrast, T_2, equals the relative projection error exactly in the long-wave limit, because there the kernel that maps a strain source to the strain it scatters is local: the once-scattered strain is the profile itself, and a cell holds it as the profile's projection (§8). In three dimensions the same kernel has, besides its local part, a part that falls as the inverse cube of the distance. The once-scattered strain then depends on how the residual of the projection is arranged in space, and cubic cells arrange it along their axes. The question is what replaces the layer's equality.

The second-order term of the scheme. Take a stiffness contrast $f(\mathbf{x})\,\Delta\mathbf{C}$, with f the profile, struck by a uniform strain $\boldsymbol{\varepsilon}_{\text{in}}$ and observed through a uniform strain $\boldsymbol{\varepsilon}_{\text{out}}$ of the outgoing P or S wave (the long-wave limit, in which the derivation is made; the test below is at $k_S a = 0.125$). The second term is the quadratic form $T_2 = \langle f, Kf\rangle$ of the operator K whose Fourier multiplier is

$$m(\hat{\boldsymbol{\xi}}) = \mathbf{a} : \boldsymbol{\Gamma}(\hat{\boldsymbol{\xi}}) : \mathbf{b}, \qquad \mathbf{a} = \Delta\mathbf{C} : \boldsymbol{\varepsilon}_{\text{out}}, \quad \mathbf{b} = \Delta\mathbf{C} : \boldsymbol{\varepsilon}_{\text{in}}, \tag{30}$$

with $\boldsymbol{\Gamma}$ the static strain Green operator of the background, which depends only on the direction $\hat{\boldsymbol{\xi}}$ of the wavevector: $m = \hat{\boldsymbol{\xi}} \cdot \frac{1}{2}(\mathbf{ab} + \mathbf{ba}) \cdot \hat{\boldsymbol{\xi}}/\mu - \frac{\lambda+\mu}{\mu(\lambda+2\mu)}(\hat{\boldsymbol{\xi}} \cdot \mathbf{a} \cdot \hat{\boldsymbol{\xi}})(\hat{\boldsymbol{\xi}} \cdot \mathbf{b} \cdot \hat{\boldsymbol{\xi}})$. Cells whose field and contrast are of the same degree compute $\langle \Pi f, K\Pi f\rangle$, Π the projection of §11.3, so with $g = f - \Pi f$ the residual,

$$\frac{T_2 - T_2^{\text{scheme}}}{T_2} = \mathscr{E} F, \qquad F = \frac{2\langle g, Kf\rangle - \langle g, Kg\rangle}{\langle f, Kf\rangle \mathscr{E}}. \tag{31}$$

In the layer $F = 1$. For a body whose spectrum is the same in every direction, as a radial profile's is, $\langle f, Kf\rangle = \bar{m}\|f\|^2$ with $\bar{m} = \operatorname{tr}\big(\frac{1}{2}(\mathbf{ab} + \mathbf{ba})\big)/(3\mu) - \frac{\lambda+\mu}{\mu(\lambda+2\mu)}\big(\operatorname{tr}\mathbf{a}\operatorname{tr}\mathbf{b} + 2\,\mathbf{a} : \mathbf{b}\big)/15$, the average of m over directions. The two remaining forms are evaluated as the cells shrink, from what the

residual looks like inside one cell.

Constant cells. In a cell of half-width h the residual is $\nabla f \cdot (\mathbf{x} - \mathbf{x}_\ell)$, a sum of three sawtooths, one along each axis. Two facts settle the two forms. Its first moments and its energy carry the same constant, $h^2/3$, and its second moments vanish, so its transform at small wavenumbers is $(h^2/3)|\xi|^2 \hat{f}$, the same in every direction, and $2\langle g, Kf\rangle \to 2\bar{m}\|g\|^2$. And a sawtooth has no mean and its harmonics lie on its own axis, carrying all its energy, so $\langle g, Kg\rangle \to m_{\mathrm{ax}}\|g\|^2$, with m_{ax} the mean of m over the three cube axes. Hence

$$F_0 = 2 - \frac{m_{\mathrm{ax}}}{\bar{m}}. \tag{32}$$

The cell enters through its axes: a residual arranged along them sees the operator along them, while the body sees it averaged over all directions.

Degree-one cells. The residual is the quadratic part of f less its projection, $\frac{1}{2}\sum_i H_{ii}(x_i^2 - h^2/3) + \sum_{i<j} H_{ij}x_i x_j$ with H the second derivatives of f. Its energy is $\sum_i H_{ii}^2 h^4/45 + \sum_{i<j} H_{ij}^2 h^4/9$ per unit volume, and its second moments, $2h^4/45$ on the diagonal and $h^4/9$ off it, give its transform at small wavenumbers as $h^4 W(\xi)\hat{f}$ with $W = \sum_k \xi_k^4/45 + \sum_{k<l} \xi_k^2\xi_l^2/9$, which is no longer the same in every direction. The parabolas $x_i^2 - h^2/3$ have their spectrum on their axes; each product $x_i x_j$ of two sawtooths has its spectrum on the lattice of the (i, j) plane, at the points (m, n), $m, n \neq 0$, with weights $1/(m^2 n^2)$, over which the averages of $u^2 v^2$ and u^4 (u, v the direction cosines) are $6G/\pi^2 - \frac{2}{5}$ and $\frac{9}{10} - 6G/\pi^2$, G being Catalan's constant, from the lattice sum $\sum_{m,n\neq 0}(m^2+n^2)^{-2} = 4\zeta(2)\beta(2) - 4\zeta(4)$. For a radial profile

$$F_1 = \frac{2\langle Wm\rangle - \frac{1}{225}\sum_i m(\mathbf{e}_i) - \frac{1}{135}\sum_{i<j} M_{ij}}{\frac{8}{225}\bar{m}}, \tag{33}$$

with $\langle Wm\rangle$ the average of Wm over directions and M_{ij} the average of m over the lattice of the (i, j) plane. A constant m gives $F_0 = F_1 = 1$, and so does a layer.

Closed forms. For an incident P wave and an outgoing SV wave the factors depend on the background alone, not on the contrast or the angle:

$$F_0^{\mathrm{PS}} = \frac{6\lambda + 11\mu}{3\lambda + 8\mu}, \qquad F_1^{\mathrm{PS}} = \frac{5\left[(57\lambda + 85\mu)\pi^2 - 315G(\lambda + \mu)\right]}{28\pi^2(3\lambda + 8\mu)}, \tag{34}$$

1.516 and 1.337 for the background of the tests ($\alpha = 5000$, $\beta = 3000\,\mathrm{m\,s^{-1}}$). For an outgoing P wave the factors are ratios of expressions linear in $\cos 2\theta$, θ the scattering angle; for the contrast of the tests F_0^{PP} runs from 1.101 near forward and back scattering to 0.923 at right angles, and F_1^{PP} from 1.066 to 0.950.

The test. The derivation is symbolic, in 31 checks, from the Navier equation to the closed forms (`Mathematica/AdaptiveOctree_T2Factor.wl`). It is tested two ways on a sphere of radius $a = 10\,\mathrm{m}$ whose contrast is uniform within $a/2$ and falls smoothly to zero at the surface, with

a stiffness contrast, at $k_S a = 0.125$: by the scheme itself, whose T_2 is set against the exact one, angle by angle and wave by wave (`scripts/measure_t2_law_3d.py`); and by evaluating the two forms of Eq. (31) directly, by fast Fourier transform on a grid finer than the cells (`scripts/measure_t2_quadratic_forms.py`); the details are in Appendix A. On 8 cells the

Table 7. The factor F of Eq. (31): the scheme on 8 cells across, the direct evaluation on finer grids (32 cells for constant, 24 for degree one), and the limit, Eqs. (32)–(34).

	constant cells			degree-one cells		
outgoing wave	scheme	direct	limit	scheme	direct	limit
P at 0.2 rad	1.074	1.092	1.101	1.073	1.067	1.066
P at $\pi/2$	0.944	0.930	0.923	0.948	0.949	0.950
SV at 0.2 rad	1.376	1.470	1.516	1.356	1.342	1.337

scheme and the direct evaluation agree to 1% where both are computed on the same grid, which confirms Eq. (31). As the cells shrink both approach the limit: for constant cells $\langle g, Kg\rangle/(\bar{m}\|g\|^2)$ goes 0.926, 0.909, 0.902 on 8, 16 and 32 cells against its limit $m_{\mathrm{ax}}/\bar{m} = 0.899$ (outgoing P at 0.2 rad), slowly, the distance falling as about $h^{1.7}$; for degree-one cells the factor is within 0.4% of the limit by 24 cells. The scheme on 8 cells is three quarters of the way to the limit for constant cells.

What it means for the second term. The second term of the error is the nonlinear part of the response times the relative projection error times F, a factor of order one that depends on the background, the direction of scattering and the type of wave, and on the cells through their axes, not on the grid. For this background and contrast it lies between 0.92 and 1.52 for constant cells and between 0.95 and 1.34 for degree-one cells, and it is one in a layer, which is why the layer's law is an equality. The constant of Table 6, formed from the largest error over the angles and with every order above the first included, is 0.96 to 1.05 times ν, inside that range. The second term is therefore known before the solve to within F, and in closed form once the direction of observation is chosen. The derivation holds for the term of second order, in the long-wave limit, for a body whose spectrum is isotropic; the terms of higher order are measured, not derived.

11.5 The nonlinear fraction without a solve

Why it is needed. The second term of the error is $\nu \mathscr{E} F$, and of its three factors only the nonlinear fraction $\nu = \max|u - T_1|/\max|u|$ (the maxima over the angles of observation) would need a solve: one solve on a coarse grid that resolves the medium, taking $|u - T_1|/|u|$ of that grid. That estimate is low, by up to a quarter for constant cells (Table 8), and the reason is the second term itself: the coarse grid's nonlinear part is low by its own relative projection error.

The method. The scheme's response is a series in the contrast, $\psi = \sum_n \psi_n$ with $M\psi_1 = b$ and $M\psi_{n+1} = KE\,\psi_n$, M the Gram matrices of the cells, b the moments of the incident wave and KE the coupling times the contrast (Eq. (26)). The first term is the Born term of §11.1; each further term costs one product with the assembled matrix and no solve. Two facts make a few terms

enough. On the exact solutions the terms of second and third order carry the nonlinear fraction to 0.2% (on the three bodies of Table 8; T_2 alone to 4%), because $|T_3|/|T_2|$ is 0.03 to 0.04. And on the coarse grid the nonlinear part of the scheme is low by its relative projection error $\mathscr{E}$, which is known before the solve (§11.3). The estimate is therefore

$$\nu \approx \frac{\max|T_2 + T_3|/(1-\mathscr{E})}{\max\left|T_1 + (T_2+T_3)/(1-\mathscr{E})\right|}, \tag{35}$$

with T_n the far fields of the scheme's terms on the coarse grid and the factor F of §11.4 taken as one.

The result. Table 8 sets the estimates against the exact nonlinear fraction on three bodies, for constant and degree-one cells. The coarse grid resolves the medium: it is refined where a cell's share of the projection error exceeds 10^{-2}, starting from eight cells (`scripts/measure_octree_nu_estimate.py`; Appendix A). The estimate of Eq. (35) is within 1.4% of the exact

Table 8. The nonlinear fraction estimated on a coarse grid, over its exact value. "Solve" is the earlier estimate (one dense solve); Eq. (35) uses two products with the matrix and the grid's projection error $\mathscr{E}$.

body	cells	number	$\mathscr{E}$	exact ν	solve	$T_2/(1-\mathscr{E})$	Eq. (35)
thin shell, $k_S a = 0.5$	constant	64	0.264	0.0151	0.746	1.047	1.014
	degree one	64	0.061		0.940	1.039	1.000
feature $0.2a$–$0.5a$, $k_S a = 1$	constant	120	0.175	0.0107	0.816	1.017	0.990
	degree one	120	0.036		0.963	1.028	0.999
feature $0.1a$–$0.3a$, $k_S a = 1$	constant	176	0.145	0.0059	0.854	1.025	1.000
	degree one	176	0.015		0.985	1.027	0.999

value in all six cases; the solve is 25% low at worst, and the second-order term alone, corrected, is 2% to 5% high, which is the share of the third-order term.

What it means. Both terms of the error are now known before the solve, without any solve: the first from one pass over the cells, the second from the relative projection error, the factor F in closed form, and the nonlinear fraction from two products with the matrix of a coarse grid. The estimate rests on the law of §11.3 applied to the coarse grid, and has been tested on three bodies at two frequencies.

11.6 The terms of third and higher order

Why they matter. The law of §11.3 was measured on the whole nonlinear part, $u - T_1$, and §11.4 derived the factor of its second-order term only. Each further term carries an error of its own, and in the layer the whole nonlinear part does not follow the second-order law: all orders together give 0.90 to 0.94 times $\mathscr{E}$ where T_2 alone gives one (Appendix F).

The layer, exactly. In the long-wave limit the layer's kernel is local, the strain scattered by a stiffness contrast being $c\,f\varepsilon$ with $c = -\Delta M/M_P$, $M_P = \lambda + 2\mu$. With field and contrast of the

same degree in each cell the scheme's term of order n is then $c^{n-1}\int(\Pi f)^n\,\mathrm{d}z$ against the exact $c^{n-1}\int f^n\,\mathrm{d}z$, for $n \le 3$ at every degree and for every n in constant cells, so the relative error of each term is a property of the profile,

$$e_n = \frac{\int f^n\,\mathrm{d}z - \int(\Pi f)^n\,\mathrm{d}z}{\int f^n\,\mathrm{d}z}, \qquad e_2 = \mathscr{E}. \tag{36}$$

Summed, the exact strain is $\varepsilon_0/(1-cf)$, and the scheme's is the Galerkin solution of $\varepsilon = \varepsilon_0 + c\,f\varepsilon$ in each cell separately, on that cell's polynomials: the error of the whole nonlinear part is the single site's, in closed form. On the heterogeneous layer of §8 (profile $1 + \frac{1}{2}\sin(2\pi z/D)$) at $\omega = 30\,\mathrm{rad\,s^{-1}}$ the scheme's T_3 has the error e_3 of Eq. (36) to 10^{-4}, for constant, degree-one and degree-two cells on 4 to 16 cells, in reflection and in transmission, and to 0.5% at $\omega = 300\,\mathrm{rad\,s^{-1}}$; $e_3/\mathscr{E} = 2.455$ on every grid, which is $3\langle f\rangle_{g^2}\int f^2\,\mathrm{d}z/\int f^3\,\mathrm{d}z$ with $\langle f\rangle_{g^2}$ the mean of the profile weighted by the squared residual. Because $T_3/T_2 = -0.078$, the second and third terms together have 0.877 times $\mathscr{E}$, and the whole nonlinear part at the full contrast 0.896 times $\mathscr{E}$, which the cell-by-cell Galerkin solution reproduces to four digits with $c = -0.0640$ from the contrast; the exact layer's T_3/T_2 returns the same c to five digits (`scripts/measure_layer_t3_law.py`). This is the 0.90 of Appendix F.

Three dimensions. The kernel is not local, and the third-order term involves the once-scattered strain fields themselves: with $\mathbf{u_b} = \boldsymbol{\Gamma} * (f\mathbf{b})$ and $\mathbf{u_a} = \boldsymbol{\Gamma} * (f\mathbf{a})$ in the notation of Eq. (30), $T_3 = \int f\,\mathbf{u_a} : \Delta\mathbf{C} : \mathbf{u_b}\,\mathrm{d}V$, and constant cells compute $\int \Pi f\,\Pi\mathbf{U_a} : \Delta\mathbf{C} : \Pi\mathbf{U_b}\,\mathrm{d}V$ with $\mathbf{U} = \boldsymbol{\Gamma} * (\Pi f\,\cdot)$. Its factor F_3, defined as F is by Eq. (31), therefore depends on the body's own field and not only on averages of m.

The closed form. The difference is expanded to second order in the half-width, as for the second term, with $\mathbf{u} = \Pi\mathbf{u} + \mathbf{u}'$ the cell decomposition of each field. Four pieces appear, each from a lemma already in hand. The residual g meets a smooth field $w_{\mathbf{ab}} = \mathbf{a} : \boldsymbol{\Gamma} * (f\,\Delta\mathbf{C} : \mathbf{u_b})$ through its first moments, giving $(h^2/3)\nabla f \cdot \nabla w_{\mathbf{ab}}$. The field scattered by the residual is a sawtooth on each cube axis carrying $\boldsymbol{\Gamma}$ on that axis, $\mathbf{G}_{i\mathbf{b}} = \boldsymbol{\Gamma}(\mathbf{e}_i)\mathbf{b}$, with amplitude $\partial_i f$. The smooth fields vary across a cell as sawtooths of amplitude $\partial_i\mathbf{u}$. And a cell's mean of a product of three fields splits exactly into the product of the means and the means of products of the departures. The terms in $(\partial_i f)^2\mathbf{GG}$ cancel, and

$$\begin{aligned}\frac{T_3 - T_3^{\text{scheme}}}{h^2/3} = \int\Big[&\nabla f \cdot \nabla(w_{\mathbf{ab}} + w_{\mathbf{ba}}) \\ &+ f\sum_i\big(\partial_i\mathbf{u_a} : \Delta\mathbf{C} : \partial_i\mathbf{u_b} - \partial_i f\,(\partial_i\mathbf{u_a} : \Delta\mathbf{C} : \mathbf{G}_{i\mathbf{b}} + \mathbf{G}_{i\mathbf{a}} : \Delta\mathbf{C} : \partial_i\mathbf{u_b})\big) \\ &+ \sum_i \partial_i f\,\big(\mathbf{u_a} : \Delta\mathbf{C} : \partial_i\mathbf{u_b} + \partial_i\mathbf{u_a} : \Delta\mathbf{C} : \mathbf{u_b}\big) \\ &- \sum_i (\partial_i f)^2\big(\mathbf{u_a} : \Delta\mathbf{C} : \mathbf{G}_{i\mathbf{b}} + \mathbf{G}_{i\mathbf{a}} : \Delta\mathbf{C} : \mathbf{u_b}\big)\Big]\,\mathrm{d}V,\end{aligned} \tag{37}$$

with $F_3 = (T_3 - T_3^{\text{scheme}})/(T_3\mathscr{E})$ and $\mathscr{E} = (h^2/3)\int|\nabla f|^2\mathrm{d}V/\int f^2\mathrm{d}V$. It is a closed form in the sense of Eq. (32): the cells enter only through $\mathbf{\Gamma}$ on their three axes, and everything else is an integral of the body's smooth, once-scattered static fields, so it needs no grid of cells. In a layer every field is local and the integrand is $3ff'^2\,\mathbf{G_a} : \Delta\mathbf{C} : \mathbf{G_b}$, which is the layer's e_3; the same expansion of the second term gives Eq. (32). The algebra is verified symbolically in 7 checks (`Mathematica/AdaptiveOctree_F3.wl`).

The test. Equation (37) is evaluated on the sphere of §11.4 by fast Fourier transform of the smooth fields alone, converged in resolution and in the padding against the images of the periodic box (`scripts/measure_f3_closed_form.py`; its value of F for the second term is 1.1016 against the exact 1.1009). It is set against two evaluations on cells: the scheme, whose terms are those of its Born series (`scripts/measure_t3_law_3d.py`), and the cell forms computed directly with $\mathbf{\Gamma}$ applied by fast Fourier transform (`scripts/measure_t3_tensor_forms.py`). On 4 cells across the last two agree to 0.3% for the outgoing P wave. The details of every measurement are in Appendix A.

Degree-one cells. With the field and the contrast in $1, x_1, x_2, x_3$ the projection on the outgoing side can be moved onto the profile, and the scheme's term is $\int \Pi f\,\mathbf{U_a} : \Delta\mathbf{C} : \Pi\mathbf{U_b}\,\mathrm{d}V$, with one projection, on the incident side; for constant cells it is the form above. The residual is now the quadratic part of the profile less its projection, $\frac{1}{2}\sum_i f_{,ii}P_i + \sum_{i<j} f_{,ij}Q_{ij}$ with $P_i = x_i^2 - h^2/3$ and $Q_{ij} = x_ix_j$ (local coordinates), and every term reduces to two cell rules. A cell-scale function meets a smooth one through their second derivatives,

$$B(S,T) = h^4\Big[\sum_i \frac{S_{,ii}T_{,ii}}{45} + \sum_{i<j}\frac{S_{,ij}T_{,ij}}{9}\Big], \tag{38}$$

and the field the residual scatters carries $\mathbf{\Gamma}$ on the axes for its parabolas and $\mathbf{M}_{ij}$, the average of $\mathbf{\Gamma}$ over the lattice of the (i,j) plane with $\langle u^2v^2\rangle = 6G/\pi^2 - \frac{2}{5}$ (§11.4), for its products. The term in two scattered residual fields is absent, because the projection removes the part of $\mathbf{U_b}$ at the scale of the cell, and

$$\begin{aligned} T_3 - T_3^{\text{scheme}} = \int\Big[&B(f\Delta\mathbf{C}\mathbf{u_a}, \mathbf{u_b}) + B(f, w_{\mathbf{ab}} + w_{\mathbf{ba}} + \mathbf{u_a} : \Delta\mathbf{C} : \mathbf{u_b}) - H(f\Delta\mathbf{C}\mathbf{u_a}, f; \mathbf{b}) \\ &- fK(f; \mathbf{a}, \mathbf{u_b}) - \mathbf{u_a} : \Delta\mathbf{C} : B(f, \mathbf{u_b}) - K_2(f; \mathbf{a}) : \Delta\mathbf{C} : \mathbf{u_b}\Big]\mathrm{d}V, \end{aligned} \tag{39}$$

where H, K and K_2 are B with one smooth factor replaced by the scattered residual,

$$\begin{aligned} H(S, f; \mathbf{b}) &= h^4\Big[\sum_i \frac{S_{,ii} : \mathbf{G}_{i\mathbf{b}}\, f_{,ii}}{45} + \sum_{i<j}\frac{S_{,ij} : \mathbf{M}_{ij}\mathbf{b}\, f_{,ij}}{9}\Big], \\ K(f; \mathbf{a}, \mathbf{u}) &= h^4\Big[\sum_i \frac{f_{,ii}\,\mathbf{G}_{i\mathbf{a}} : \Delta\mathbf{C} : \mathbf{u}_{,ii}}{45} + \sum_{i<j}\frac{f_{,ij}\,\mathbf{M}_{ij}\mathbf{a} : \Delta\mathbf{C} : \mathbf{u}_{,ij}}{9}\Big], \\ K_2(f; \mathbf{a}) &= h^4\Big[\sum_i \frac{f_{,ii}^2\,\mathbf{G}_{i\mathbf{a}}}{45} + \sum_{i<j}\frac{f_{,ij}^2\,\mathbf{M}_{ij}\mathbf{a}}{9}\Big], \end{aligned}$$

and $\mathscr{E} = h^4 \int [\sum_i f_{,ii}^2/45 + \sum_{i<j} f_{,ij}^2/9] \mathrm{d}V / \int f^2 \mathrm{d}V$. In a layer the integrand is $(h^4/45)\,\kappa\,(6f'^2 f'' + 3ff''^2)$, $\kappa = \mathbf{G_a} : \Delta\mathbf{C} : \mathbf{G_b}$, which is the direct expansion of $\int f^3 - \int (\Pi f)^3$; and the same rules give the second term as $2B(f, \mathbf{a} : \mathbf{u_b}) - \mathbf{a} : K_2(f; \mathbf{b})$, whose ratio is the F_1 of Eq. (33). The algebra is verified in 8 checks (`Mathematica/AdaptiveOctree_F3Linear.wl`). The closed form gives $F_3 = 1.475$, 1.428 and 1.387 for the outgoing P wave at 0.2, 0.885 and $\pi/2$ rad and 1.892 for the SV wave, and its value of F_1 for the second term is 1.0672 against the exact 1.0659. The direct evaluation on cells approaches it at second order in the cell size, and extrapolated from 16 and 24 cells gives 1.475, 1.436, 1.404 and 1.874. The scheme on 8 cells gives 1.350, 1.299, 1.254 and 1.79, which the direct evaluation on 8 cells reproduces (1.351, 1.264 and 1.79): for degree-one cells F_3, unlike F_1, is still far from its limit on 8 cells.

Table 9. The factors of the second and third terms and of the whole nonlinear part, $k_S a = 0.125$: the scheme on 8 cells across, the direct evaluation of F_3 (constant cells: 32 cells, outgoing P, the SV wave not converged; degree-one cells: extrapolated from 16 and 24), and the closed forms, Eqs. (37) and (39).

outgoing wave	F (scheme)	F_3 (scheme)	F_3 (direct)	F_3 (closed form)	T_3/T_2	nonlinear part
constant cells						
P at 0.2 rad	1.073	1.600	1.586	1.594	−0.036	1.054
P at $\pi/2$	0.944	1.494	1.417	1.410	−0.042	0.921
SV at 0.2 rad	1.375	2.145		2.46	−0.021	1.358
degree-one cells						
P at 0.2 rad	1.066	1.350	1.475	1.475	−0.036	1.056
P at $\pi/2$	0.948	1.254	1.404	1.387	−0.042	0.935
SV at 0.2 rad	1.341	1.79	1.874	1.892	−0.021	1.332

The direct evaluation converges as the cells shrink (1.62, 1.58, 1.59 on 8, 16 and 32 cells, outgoing P at 0.2 rad) to the closed form, 1.594, and agrees with it to 0.5% at both angles. The scheme on 8 cells is three quarters of the way to it for the SV wave, as it is for the second term. The nonlinear part's factor is the mean of the terms' factors weighted by the terms: $(F + F_3 T_3/T_2)/(1 + T_3/T_2) = 1.054$ against the measured 1.054, so the fourth and higher terms contribute 0.05%.

What it means. Every order has its own factor, larger than the second's, and the factor of the nonlinear part is their mean weighted by the terms. Where the contrast is moderate the second term dominates ($|T_3/T_2|$ is 0.02 to 0.04 here) and the nonlinear part's factor is F to within $|T_3/T_2|(F_3 - F)$, two percent here; the estimate of §11.5 already carries the third term. In the layer the whole series is derived; in three dimensions the factor of the third term is derived in closed form for constant and for degree-one cells, Eqs. (37) and (39), and those of the higher terms are measured.

12 Matching the bases

A smooth medium: the degrees must be matched. The test body is a sphere of radius a whose contrast is a weak halo, falling smoothly from 5% of the central value to zero across the whole radius, plus a compact feature in which it falls from the central value to the halo

between $0.1a$ and $0.3a$, at $k_S a = 1$ (Appendix A). Hold the contrast constant in every cell and raise the field to degree one. The error rises, at four times the unknowns: from 7.15×10^{-3} to 8.31×10^{-3} on the uniform grid of 408 cells. The two terms, computed without a solve, predict it (1.11×10^{-2} against 1.34×10^{-2}). On a smoothly varying medium the first-order error of a cell with a constant contrast is not the wave's: it is the error of holding the contrast constant, which a richer field cannot repair, and with a constant field the wave's error partly cancels it. This is the rule of §8 that the two bases must be matched, now with the error predicted before the solve.

A medium that is piecewise constant. Where the medium is constant on each cell, a constant contrast is exact, the second term vanishes, and the degree is better spent on the field. That needs cells whose faces follow the interfaces of the medium; where an interface falls inside a cell, a richer field is worse than a constant one, as on the smooth body, and for the scalar problem Anderson et al. [51] likewise find that high order is lost at jumps of the medium unless the grid follows them.

13 Discussion

Figure 6 collects the convergence of every test, with the order fitted to each by least squares (`scripts/plot_convergence_orders.py`).

What each basis is for. A voxel holds two functions, and the results give each basis its part. The basis of the wavefield sets how well the voxel carries the field across itself: in a uniform medium the scheme converges at order $2p + 2$ with the constant c_p of Eq. (20), for P and S waves and at normal and oblique incidence alike. The basis of the medium sets how well the voxel *is* the medium: with degree r the order is capped at $2r + 2$ (Eq. (22)). Neither can make up for the other, and the second-order term shows why. The field scattered once is a copy of the medium's profile, so the wavefield basis must carry the profile as well as the wave, and the error of that term is the relative projection error of the profile at the lower of the two degrees (Eq. (24)). At the frequencies of interest the wave varies little across a cell and the medium may vary much, so it is the medium, not the wavelength, that decides the degree and the size of the cells. Two cases bracket the choice. Where the medium is piecewise constant on the cells, as in a layered model with voxel faces on its interfaces, a constant is exact for the medium, the projection error vanishes, and the degree of the field is spent entirely on the wave. Where the medium varies smoothly within the cells, the two degrees should be equal: a higher degree for the contrast than for the field is wasted, and a higher degree for the field improves only the Born term. That the two degrees can be chosen independently is the practice of higher-order electromagnetic volume methods [1]; what the present results add is the rule for choosing them, and the error each choice leaves.

Where the error of a voxel scheme lives. The results separate the two approximations a voxel scheme makes. The *coupling* between voxels makes none: averaged over the source cell it is the continuum's cell integral exactly, between planes by the form factor's zeros and within a

plane by the tiling identity, and the self term it omits is exactly the one the single site carries. Every error therefore comes from the *single site*: from the internal field it assumes and, where the medium varies within the cell, from the medium it is given. Its T-matrix sums the rescattering within the voxel exactly, but for the voxel as represented, and the split of the second-order term in §8 measures the consequence: the error sits in the part whose two scatterings lie in one cell, and the part between cells carries at most 0.2% of it. For a uniform field that error is the midpoint rule's, known in closed form; for a linear field it is the next term of the same expansion. The converse is the practical warning. A kernel that is not the cell integral (the point Green's tensor, or a cell average that does not match the single site's closure) leaves a local term that does not vanish as the voxels shrink (§4): the evanescent lattice orders carry a static contribution of fixed size per voxel volume. A scheme built that way converges to the wrong limit, and refinement cannot reveal it, because the error is scale-invariant.

Relation to other methods. The partition of the error is the coupled-dipole method's choice of polarisability seen from the other side: for space-filling cubes the tiling identity fixes the local field exactly, and the polarisability is neither needed nor a free parameter. The voxel of degree p is close kin to the spectral element of seismology; the two share a Galerkin condition on polynomial bases, and differ as a differential formulation from an integral one, and so suit different problems. Appendix G sets out both relations, and where each method is the better choice.

The tension between the two error sources. The coupled-dipole method has two sources of error, the single site and the coupling between sites, and each has been corrected on its own: the polarisability by the Clausius–Mossotti relation, the lattice dispersion relation or the exact static limit, and the coupling by integrating or filtering the Green tensor. The two are not independent. The point kernel's evanescent orders carry a static local term whose size per voxel volume does not shrink with the voxels (§4), and the single site must cancel exactly that term. A polarisability tuned for one kernel is therefore wrong for another: the lattice dispersion relation corrects the point-dipole lattice, and changing the kernel to an integrated or a filtered one changes the term it must cancel. Tuning either alone moves the error between the two, and refinement cannot reveal a mismatch, because the surviving term is scale-invariant. For space-filling cubes the tension is resolved rather than balanced: the cell-averaged kernel omits exactly the self term that the closure carries (Eq. (15)), so the coupling error vanishes identically and every remaining error is the single site's, the variation of the field across the cell. That error has a second tension of its own. It is a midpoint-rule error whose constants have opposite signs in reflection and transmission (Eq. (16)), so any scalar adjustment of the single site that improves one worsens the other; the package's far-field corrections reduce the reflection error by a factor 4.45 and increase the transmission error by 1.7. No polarisability, however chosen, can remove it. What removes it is a richer internal field: the first moment lowers the error by two orders in kd in both observables at once.

The cost of higher order, and a fast solver. A first-moment voxel carries four times as many unknowns as a uniform one, but by the closed-form constants it reaches a reflection error of 10^{-6} with voxels 58 times larger in each direction; the stratified model of §8 measures it directly, 347 planes of uniform-field voxels against 8 first-moment voxels. And because the discretisation error is known in advance and the coupling between cells is exact, a fast product need be no more accurate than that error, and can be budgeted separately from it. Appendix H sets out the cost, and what the error analysis offers a fast solver.

Limitations and outlook. The results concern laterally uniform media, a homogeneous layer and a randomly stratified one, for which the discrete system is exactly a chain of planes; a laterally heterogeneous medium needs the first-moment voxel in a full three-dimensional solver, where the gradient channels couple through the evanescent lattice orders (§7) and the tiling identity holds only to $O(k_\parallel h)$. The smoothly graded sphere of §10 shows the mean-only voxel converging at second order on a laterally heterogeneous body against an exact solution, and the first-moment voxel converges at fourth order there [46]. A contrast that varies within a cell, which §8 shows is needed beyond second order in a smoothly varying medium, has a further consequence in three dimensions: its linear part is odd about the cell's centre, so it couples the single site's uniform-field and first-moment blocks, which parity keeps apart for a homogeneous cube [5]. A graded voxel therefore needs at least the first-moment block; that voxel is the degree-one cell of Part II (§9). The numerical evidence is at moderate, physically typical contrasts (up to 15% in density and moduli) and at $k_S h \le 0.3$, the range in which the single-site T-matrix is validated; the closed forms are first order in the contrast, and the departures from them at the full contrast are of second order. What is established is the *order* of convergence and its leading constant, not the behaviour at strong contrast, where the single site's own closure may limit the accuracy before the lattice coupling does. The single site's dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§6–7). Two extensions follow directly. The first is a fast three-dimensional solver. The second is a stratified *background*, a layered reference medium, as distinct from the stratified contrast of §8. For a laterally uniform layer of voxels the orders, the first-order factor and the projection-error law all survive it, at normal and at oblique incidence, for P, SV and SH waves, in a background of two half-spaces and in one of three layers (§8). What remains is a body of finite lateral extent in such a background, where the image of a voxel in an interface it touches is singular.

14 What is established in three dimensions, and what is not

- *Established twice.* The wave factor, Eqs. (27) and (28): a symbolic derivation and an independent numerical implementation, and both against the solver.
- *Established against exact solutions, by one implementation.* The two-term separation (Table 6) and the matched-degree rule (§12): one frequency each, two bodies, both spheres with radial profiles.
- *Derived, and established twice.* The factor of the second term, Eqs. (31) to (34): a symbolic

derivation, and the scheme and a direct evaluation against it (Table 7). It is derived for the term of second order, in the long-wave limit, for a body with an isotropic spectrum.

- *Established against exact solutions, by one implementation.* The nonlinear fraction from two products with the matrix of a coarse grid, Eq. (35): within 1.4% on three bodies, for constant and degree-one cells (Table 8).
- *Derived in the layer; measured twice in three dimensions.* The terms of third and higher order (§11.6): in the layer each term's error and the whole series in closed form, the latter the cell-by-cell Galerkin solution of the local problem; in three dimensions the third term's factor for constant cells in closed form, Eq. (37), verified symbolically and against the scheme and a direct evaluation, and for degree-one cells, Eq. (39), likewise. The factors of the fourth and higher terms are not derived.
- *Size.* The solver is dense and stops near 20 000 unknowns, so the bodies are small: up to 1256 constant cells or 408 degree-one cells.
- *Earlier work.* The survey behind §1 found no statement of the two-term law, of the cell's factor or of the matched-degree rule, and no elastic volume scheme with polynomial cells. It rests on some twenty searches and on the opening pages of most of the works cited.

15 Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, first by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree p and the contrast to degree r in each voxel, the scheme converges at order $\min(2p+2, 2r+2)$, with the leading error $c_p(kd)^{2p+2}$ in closed form (Eq. (20)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree $\min(p, r)$: it is the single site's, whose T-matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

In three dimensions the error of the scattered field is two terms, and both are known before the solve. The first is the error of the term of first order in the contrast, from one pass over the cells and in closed form for a cell of uniform contrast; its one-dimensional case is the constants c_p of the layer. The second is the nonlinear fraction of the response times the relative projection error of the medium, times a factor of order one that is one in a layer and in three dimensions depends, in closed form, on the background and the direction of scattering through the axes of the cells. The nonlinear fraction comes from two products with the matrix of a coarse grid, so neither term needs a solve, and the term of third order has a closed-form factor of its own.

The error is known in advance. For every discretisation studied here the error is not something to be discovered by refining until the answer stops changing. Its order and its leading constant are known before anything is computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the collocation voxel (Eq. (16)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the voxels of degree 0, 1 and 2 (Eq. (20)), and the order $\min(2p+2, 2r+2)$ where the medium varies within a cell (Eq. (22)), with the size of the error there given by the relative projection error of the medium (Eq. (24)), which is computed from the model and the grid alone. That is worth more than the order alone, for three reasons. First, the cell size and the degree of the voxel that a required accuracy needs follow from the wavenumber and the medium, before the computation and without a refinement study. Secondly, it turns a refinement study from a hope into a test. A scheme of this kind can converge smoothly to the wrong limit (§4), and smooth convergence alone cannot reveal it; but the difference between two grids must have the known order and the known constant, and a computation whose differences do not is in error from some cause other than the discretisation. Thirdly, it shows where effort pays. The error is that of representing the field and the medium within a cell, so raising the degree of the cell does what multiplying the cells cannot: in the stratified model one first-moment voxel per layer reaches 10^{-6} where the uniform-field voxel needs about forty. For a body with three-dimensional structure the error is known in advance in the same way, through its two terms (§11), which also tell a grid of unequal cells where its cells are needed and give a fast solver the tolerance it must meet.

The cells, their blocks and the solver are `cubic_scattering/graded_voxel/octree.py`, with tests in `cubic_scattering/tests/test_graded_voxel_octree.py`. The wave factor is derived in `Mathematica/AdaptiveOctree_WaveFactor.wl` and measured by `scripts/measure_octree_wave_term.py`; the factor of the second term is derived in `Mathematica/AdaptiveOctree_T2Factor.wl` and measured by `scripts/measure_t2_law_3d.py` and `scripts/measure_t2_quadratic_forms.py`; Table 8 is `scripts/measure_octree_nu_estimate.py`; the terms of higher order are `scripts/measure_layer_t3_law.py`, `scripts/measure_t3_law_3d.py` and `scripts/measure_t3_tensor_forms.py`, with the closed form of the third term derived in `Mathematica/AdaptiveOctree_F3.wl` and evaluated by `scripts/measure_f3_closed_form.py`; Table 6 is `scripts/measure_octree_two_term.py`; the constant medium with a richer field is `scripts/pilot_octree_hp_refinement.py`. Their results are stored in `LatexPDFs/PolynomialVoxelError/figures`. Material on octree refinement, removed from this paper, is archived in `LatexPDFs/OctreeRefinementArchive/`.

A The experiments in detail

Every measurement of §§11.4–11.6 is set out here: the body, the background and the contrast, the waves and how they are observed, how each term of the series is extracted, what is neglected and how small it is, and the numerical parameters of each evaluation.

Background, contrast and waves. The background is isotropic with $\alpha = 5000$ and $\beta = 3000\,\mathrm{m\,s^{-1}}$ and $\rho = 2500\,\mathrm{kg\,m^{-3}}$. The contrast is $\Delta\lambda = 2\,\mathrm{GPa}$, $\Delta\mu = 1\,\mathrm{GPa}$ and $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$ times the profile, or the stiffness alone ($\Delta\rho = 0$) where the long-wave derivations of §§11.4 and 11.6 are tested, since the density enters the second and third terms only at higher order in $k_S a$. The incident wave is a P wave along axis 1. The scattered field is observed at nine angles from 0.2 to $\pi - 0.2\,\mathrm{rad}$ in the plane of axes 1 and 2, at 5×10^8 radii, where the near-field terms are negligible, and split into its P part (radial) and its S part (transverse).

The bodies. All are spheres of radius $a = 10\,\mathrm{m}$ whose contrast is a function of radius, built from the quintic step $s_5(x) = 10x^3 - 15x^4 + 6x^5$ on $[0, 1]$, so that their exact solutions are known by the radial reference of §10. The sphere of §§11.4 and 11.6 has full contrast within $a/2$ and $s_5((a-r)/(a/2))$ outside it. The thin-shell sphere of §§11.3 and 11.5 has full contrast within $0.75a$. The two bodies with a feature have a halo $0.05\,s_5((a-r)/(a-r_1))$ plus $0.95\,s_5((r_2-r)/(r_2-r_1))$, with the feature between $r_1 = 0.2a$ and $r_2 = 0.5a$, or between $0.1a$ and $0.3a$ (§§11.5 and 12).

The terms of the series. The scheme's terms are those of its Born series (`octree.born_series_octree`), each one product with the assembled matrix, on uniform grids of 4, 6 and 8 cells across (§11.4 uses the equivalent uniform solver of the voxel papers, with blocks by the cube's symmetry). The exact terms are read from the exact sphere at scaled contrasts $\pm d$ and $\pm 2d$: $T_1 = [8(u_d - u_{-d}) - (u_{2d} - u_{-2d})]/12d$, $T_2 = [16(u_d + u_{-d}) - (u_{2d} + u_{-2d})]/24d^2$ and $T_3 = [(u_{2d} - u_{-2d}) - 2(u_d - u_{-d})]/12d^3$, u_s the field at contrast s times the full one (§11.4 uses $T_2 = (u_d + u_{-d})/2d^2$ at $d = 10^{-3}$, whose error, of order $d^2 T_4/T_2$, is negligible there). Each carries a truncation error of order d^2 times the ratio of the next term of its parity, and a cancellation: T_3 is recovered from the odd part through a fraction $3d^2 T_3/T_1$ of it, about 10^{-7} at $d = 10^{-2}$. For constant cells, whose $\mathscr{E}$ is about 0.06 on 8 cells, $d = 10^{-2}$ suffices. For degree-one cells, whose $\mathscr{E}$ is 5×10^{-3}, the error to be measured in T_3 is a few parts in 10^3, and $d = 10^{-2}$ gave a factor that was not symmetric about $\pi/2$ (which the static derivation requires) and that worsened at lower frequency: the cancellation, not the physics. With $d = 0.05$, 0.1 and 0.2 the factor is symmetric, its truncation falls as d^2 (successive differences in the ratio 4), and the Richardson extrapolations from the two pairs agree to 10^{-4}; these are the values given. The frequency is $k_S a = 0.125$, where the half-width of a cell is 0.016 wavelengths and the dynamic corrections to the factors are negligible; the derivations are static.

The measured ratio. For each angle and each wave type the ratio is $\mathrm{Re}[(T - T^{\mathrm{scheme}})\cdot\bar{T}]/(|T|^2\mathscr{E})$, T the exact term as a displacement vector; for the S wave at $\pi/2$ the second and third terms vanish and the ratio is not defined. $\mathscr{E}$ is the relative projection error of the profile on the cells, by a Gauss rule of 10 points per axis in each cell (§11.4) or of 6 points on descendants three levels deep (§§11.5 and 11.6).

The direct evaluations on cells. The forms of Eq. (31) and of the third term are evaluated by fast Fourier transform with the profile sampled at s midpoints per cell and axis, on a periodic box larger than the body by a padding factor, with $\mathbf{\Gamma}$ at zero wavevector set to its average

over directions. The cell projection is the discrete one on the samples, which keeps the residual orthogonal to the cell's functions on them and converges as $1/s$; the exact L^2 projection, sampled, leaves spurious low moments in the residual that are harmless for constant cells and large for the small residual of degree-one cells. For the second term the exact $\langle f, Kf\rangle = \bar{m}\|f\|^2$ of a radial profile replaces the transform's own value, which carries a small-wavevector error of 1% to 10%. The grids run from $s = 4$ to 32 and 4 to 32 cells across, on boxes of up to 480^3 points; for degree-one cells $s = 4$ is too coarse and is not used.

The closed forms. Equations (32), (33) and (34) are evaluated exactly (`Mathematica/AdaptiveOctree_T2Factor.wl`). Equations (37) and (39) need the body's smooth static fields, which are computed by fast Fourier transform with 40 to 96 points across the body and a padding of 2 to 5; the factors are converged in resolution to four digits, and the padding, against the periodic images whose coupling through the inverse-cube kernel decays slowly, is checked by the second term, whose closed form is known: at a padding of 5 it is 1.1016 against 1.1009 for the P wave and 1.521 against 1.516 for the SV wave (`scripts/measure_f3_closed_form.py`).

The nonlinear fraction. The coarse grid is an octree, refined from $2\times2\times2$ cells while a cell's share of the projection error exceeds 10^{-2}, with cells no smaller than $a/16$ ($a/32$ for the feature between $0.1a$ and $0.3a$). Its relative projection error $\mathscr{E}$ is that of the profile on its cells at their degree. The exact nonlinear fraction is formed from the exact field and the exact Born term, the latter by a central difference at $d = 10^{-3}$ (`scripts/measure_octree_nu_estimate.py`).

The layer. The layer of §11.6 is the heterogeneous layer of §8: thickness $D = 2\,\mathrm{m}$, the profile $1 + \frac{1}{2}\sin(2\pi z/D)$, a plane source at $z = -12\,\mathrm{m}$, observers at $z = -1\,\mathrm{m}$ (reflection) and $z = 4\,\mathrm{m}$ (transmission), the contrast above with its density, normal incidence, 4, 8 and 16 cells of degree 0, 1 and 2, at $\omega = 30$ and $300\,\mathrm{rad\,s^{-1}}$. Its terms are read as above with $d = 10^{-2}$, from the scheme and from the exact layer in 30-digit arithmetic. The constant of the local kernel is $c = -\Delta M/M_P = -0.0640$, with $\Delta M = \Delta\lambda + 2\Delta\mu$ and $M_P = \lambda + 2\mu$, and the cell-by-cell Galerkin solution is computed with the Gauss rule of the cells (`scripts/measure_layer_t3_law.py`).

B Further tests of the Legendre voxel

Oblique incidence. The scheme is built spectrally. By Poisson summation the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_\parallel$ is

$$C_{ab}(i,j) = \frac{1}{d^2}\sum_{\mathbf{g}} L_a(\boldsymbol{\kappa})\,L_b(-\boldsymbol{\kappa}) \iint \phi_a(z)\,\hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z')\,\phi_b(z')\,\mathrm{d}z\,\mathrm{d}z', \qquad \boldsymbol{\kappa} = \mathbf{k}_\parallel + \mathbf{g}, \tag{40}$$

with L_a the lateral Legendre factors. The whole-space tensor decays as $1/k_z^2$ and the nine-component state adds at most two powers of k_z, so each entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma + \mathrm{i}\beta\,\mathrm{sgn}(z)/2]\,\mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \le 8$.

Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 10). The mean-only voxel is second order and the first-moment voxel fourth order in both

Table 10. Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20°, with the mean-only and first-moment voxels of Eq. (40). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_\parallel h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_\parallel = 0$ the construction reproduces Eq. (18) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (18) with $M_P \to \mu$, $k_P \to k_S$ and $\Delta M \to \Delta\mu$; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full 9×9 construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique incidence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders are those of Table 11. The orders, and with $k = k_S$

Table 11. Observed order of convergence ($n = 2 \to 8$, $\omega = 300\,\mathrm{rad\,s^{-1}}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9°$.

	mean only	mean and first moments
SV at 20°: R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30°: R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40°: R_{SH}	2.00	4.00

the closed-form constants of Eqs. (16) and (19), are those of the incident P wave.

Degree two at oblique incidence. In three dimensions the voxel of degree 2 carries, on each of the nine components of its state, the ten Legendre products of total degree at most two in (z, x, y). In the construction of Eq. (40) it converges at sixth order against the exact layer for incident P, SV and SH waves (Table 12; `scripts/measure_oblique_degree.py`). The test is run at $\omega = 3000\,\mathrm{rad\,s^{-1}}$ ($k_PD = 1.2$, $k_SD = 2$), ten times the frequency of the other tables, because at $300\,\mathrm{rad\,s^{-1}}$ a sixth-order error on these grids lies below the round-off of double precision. The exact reflection coefficient is that of Kennett's recursion, an independent algorithm, and the same computation with the bases of degree 0 and 1 returns orders 2 and 4 in all five cases. At normal incidence the error on four planes, 6.95×10^{-9}, is the $c_2(kd)^6 = 7.2\times10^{-9}$ of Eq. (20) to 4%. Doubling the number of lattice orders retained in Eq. (40) changes the errors by less than 1%. Unlike the degrees 0 and 1, this degree is implemented once at oblique incidence; its check is the exact solution.

Table 12. The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000\,\mathrm{rad\,s^{-1}}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

	error, $p = 2$			order	orders
incident wave	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0$ / $p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20°: R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20°: R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20°: R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40°: R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

The spectral construction is implemented twice. The second implementation (`scripts/crosscheck_first_moment_voxel.py`) shares only the definition of the scheme: it takes the depth dependence of the transformed kernel from an independent residue construction, obtains the local weight as a numerical large-k_z limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two depths. The raw reflected displacements of the two agree to 9×10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and 20°, $n = 1, 2, 4$, with and without the first moments. The one-dimensional scheme of any degree is also implemented twice. The second implementation (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral by Gauss quadrature, split at the receiver within a voxel, and the exact layer from its own interface system, in double precision. Its errors agree with the 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in transmission); at $\omega = 1200\,\mathrm{rad\,s^{-1}}$ it measures orders 4.02 / 3.99 and 6.02 / 6.00; for one plane as the layer thins its differences from the exact layer fall as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0$, 1 and 2; and it rederives c_p for $p \le 5$. Figure 6(a,b) shows the convergence, with the orders fitted to it.

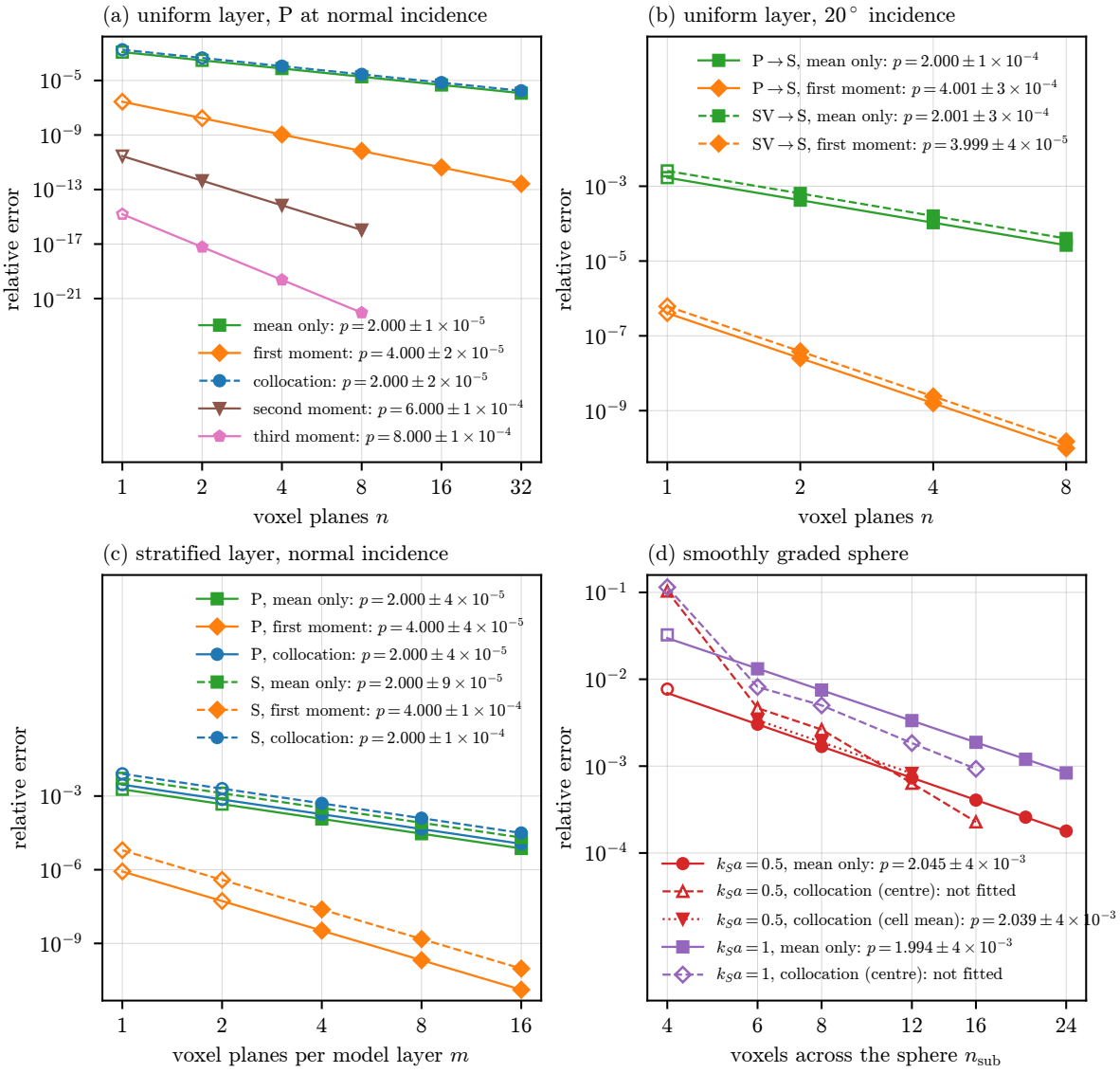


Figure 6. Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; Eq. (15)), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (18) and (40)). (a) The uniform layer, P reflection at normal incidence ($D = 2\,\mathrm{m}$, $\omega = 300\,\mathrm{rad\,s^{-1}}$, the contrast of §6), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20°, fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §8, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\mathrm{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and not fitted, since its two $O(h^2)$ errors of opposite sign leave no single order to fit; and, at $k_S a = 0.5$, the same collocation voxel with the cell mean of the contrast, dotted, which recovers second order.

Alt text: Four log–log panels of relative error against refinement, each showing data points and fitted straight lines with the fitted order in the legend. (a) Uniform layer at normal incidence, 1 to 32 voxel planes: the collocation and mean-only lines have order 2.000, the first-moment line order 4.000, falling from about 3×10^{-7} to 2×10^{-13}; the second-moment line has order 6.000 and the third-moment line order 8.000, over 1 to 8 planes, reaching about 10^{-22}. (b) Uniform layer at 20 degrees, 1 to 8 planes: P-to-S and SV-to-S lines, order 2.00 for the mean-only voxel and 4.00 for the first-moment voxel. (c) Stratified layer, 1 to 16 planes per model layer: P and S reflection, order 2.000 for collocation and mean-only, 4.000 for first moment. (d) Smoothly graded sphere, 4 to 24 voxels across: two straight falling lines of order 2 for the mean-only voxel, for $k_S a = 0.5$ and 1, from about 10^{-2} and 3×10^{-2} at 4 voxels to 2×10^{-4} and 8×10^{-4} at 24; two dashed collocation curves starting near 10^{-1} at 4 voxels, dropping steeply to 6 and then falling faster than the mean-only lines, crossing below them by 16 voxels; and a dotted line of order 2 for the cell-mean collocation voxel, close to the mean-only line.

C The stratified-background construction

The experiment is built in four steps.

The kernel. At each lateral wavenumber $\boldsymbol{\kappa}$ the whole-space kernel between two depths factorises over the plane-wave modes, three going down and three going up (P, SV, SH): $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') = \mathsf{D}_\downarrow\, \mathsf{E}(z - z')\, \mathsf{S}_\downarrow$ for $z > z'$, with $\mathsf{S}_\downarrow$ the 3×9 matrix of mode amplitudes radiated downwards by a nine-component source, $\mathsf{E}(L) = \mathrm{diag}(\mathrm{e}^{\mathrm{i}\gamma_m L})$ the propagation over a distance L with γ_m the vertical wavenumber of mode m, and $\mathsf{D}_\downarrow$ the 9×3 matrix of the modes' states; and likewise upwards. With the interfaces at z_A above and z_B below, $H = z_B - z_A$, and R_A, R_B the 3×3 reflection matrices of the two interfaces seen from inside the layer, a source at z' sends $\mathsf{s}_\downarrow = \mathsf{S}_\downarrow q$ down and $\mathsf{s}_\uparrow = \mathsf{S}_\uparrow q$ up, and the reverberation it sets up is

$$\begin{aligned} \mathsf{d}_A &= \big(\mathsf{I} - \mathsf{R}_A \mathsf{E}(H) \mathsf{R}_B \mathsf{E}(H)\big)^{-1} \mathsf{R}_A \big[\mathsf{E}(z' - z_A)\, \mathsf{s}_\uparrow + \mathsf{E}(H)\, \mathsf{R}_B\, \mathsf{E}(z_B - z')\, \mathsf{s}_\downarrow\big], \\ \mathsf{u}_B &= \mathsf{R}_B \big[\mathsf{E}(z_B - z')\, \mathsf{s}_\downarrow + \mathsf{E}(H)\, \mathsf{d}_A\big], \\ \hat{\mathbf{G}}_{\mathrm{rev}}(\boldsymbol{\kappa}; z, z')\, q &= \mathsf{D}_\downarrow \mathsf{E}(z - z_A)\, \mathsf{d}_A + \mathsf{D}_\uparrow \mathsf{E}(z_B - z)\, \mathsf{u}_B, \end{aligned} \tag{41}$$

with d_A the downgoing amplitudes leaving the upper interface and u_B the upgoing ones leaving the lower. It is added to the whole-space kernel in every block of Eq. (40), at every order of the reciprocal lattice. It is smooth and separable in z and z', so its integrals over the cells are one-dimensional moments of exponentials. For two half-spaces $\mathsf{R}_A = 0$ and only the second line survives.

The interfaces. R_A, R_B and the transmission matrices into and out of the layer follow from continuity of displacement and traction at each interface: six equations in the states of the six modes on either side, solved for each lateral wavenumber, evanescent orders included.

The incident field and the output. A plane wave of unit amplitude arrives from the upper half-space. Its field in the layer, a downgoing part and an upgoing part with the reverberation summed as in Eq. (41), is the right-hand side of the scheme. The output is the upgoing wave of the same type as the incident one, arriving at z_A from the voxels and transmitted into the upper half-space: the change that the layer of voxels makes to the same-type reflection coefficient (R_{PP}, R_{SS} or R_{SH}), a quantity that does not depend on how the waves are normalised.

The reference, and a check of the construction. The exact change is the same-type reflection coefficient of the stack with the layer, less that of the background, both from Kennett's recursion [45], an independent algorithm. Before any voxel is introduced, the background's own reflection coefficient computed from the modes and the interface matrices above agrees with Kennett's to 5×10^{-15} in all twelve cases.

It is computed by two independent algorithms. The first is invariant imbedding, the impedance march of Haines et al. [50]: the six-component state of displacement and traction obeys $\mathrm{d}\mathsf{y}/\mathrm{d}z = \mathsf{A}(z)\, \mathsf{y}$, and with the four 3×3 blocks A_{ij} of A the impedance Y, which gives the traction from the displacement, obeys the Riccati equation

$$\frac{\mathrm{d}\mathsf{Y}}{\mathrm{d}z} = \mathsf{A}_{21} + \mathsf{A}_{22}\mathsf{Y} - \mathsf{Y}\mathsf{A}_{11} - \mathsf{Y}\mathsf{A}_{12}\mathsf{Y}. \tag{42}$$

It is started below the stack with the impedance of the downgoing waves of the lower half-space, which is the radiation condition, marched upwards, exactly through the homogeneous gaps and by fourth-order Runge–Kutta with the medium sampled at the stages through the graded layer, and at the top the reflection matrix follows from continuity of displacement and traction with the modes of the upper half-space. Y stays bounded, so no growing exponential is formed. The second is the propagator: the same state is integrated across the graded layer as a 6×6 matrix, with $\mathsf{A}(z) = \mathsf{F}(z)\,\mathrm{diag}(\mathrm{i}\gamma_m)\,\mathsf{F}(z)^{-1}$ built at each depth from the plane-wave modes of the medium there, F the matrix of the six modes' displacements and tractions and γ_m their vertical wavenumbers, and the radiation conditions of the two half-spaces are imposed on the result. For the graded layer the two agree to 10^{-12} in all twelve cases with 400 steps of the march, whose error falls sixteenfold for each doubling of the steps; for a uniform layer the propagator equals Kennett's recursion to 3×10^{-14}.

Oblique incidence, and a three-layer background. At oblique incidence the test is the three-dimensional scheme of Eq. (40), with P, SV and SH waves, in two backgrounds: the two half-spaces above, and a three-layer background in which the voxels lie in a layer between a slower half-space above ($\alpha = 4200$, $\beta = 2400\,\mathrm{m\,s^{-1}}$, $\rho = 2300\,\mathrm{kg\,m^{-3}}$) and the faster one below (6500, 3700, 2800), so that the layer reverberates between two interfaces. The construction is the one set out above: the reverberation added to the whole-space kernel at every order of the reciprocal lattice, Eq. (41), the interface matrices, the incident field, and the reference, Kennett's recursion [45], against which the background's own reflection coefficient is reproduced to 5×10^{-15} in all twelve cases before any voxel is introduced.

The results are in Table 13, at $\omega = 3000\,\mathrm{rad\,s^{-1}}$ ($k_S D = 2$; at $300\,\mathrm{rad\,s^{-1}}$ the sixth-order errors lie below round-off, as in Table 12) and 20° in the upper half-space. In all thirty-six cases the order is that of the whole space, with the interfaces 3 m from the layer and with the voxels against them (`scripts/measure_oblique_stratified_background.py`). The errors are close

Table 13. The layer of voxels in a stratified background at 20°, $\omega = 3000\,\mathrm{rad\,s^{-1}}$: relative error of the change of the same-type reflection coefficient on $n = 4$ planes, and the order over the last halving ($n = 2 \to 4$), for field degrees $p = 0$, 1 and 2.

background	wave	$p = 0$	$p = 1$	$p = 2$
two half-spaces, interface 3 m below	P	6.1×10^{-3} (2.03)	2.7×10^{-6} (4.02)	2.3×10^{-9} (6.04)
	SV	1.5×10^{-2} (2.11)	2.6×10^{-5} (4.13)	3.7×10^{-8} (6.09)
	SH	1.6×10^{-2} (2.11)	2.7×10^{-5} (4.13)	4.1×10^{-8} (6.09)
two half-spaces, interface against the voxels	P	5.1×10^{-3} (2.04)	3.8×10^{-6} (4.04)	2.1×10^{-9} (6.00)
	SV	1.6×10^{-2} (2.10)	2.5×10^{-5} (4.14)	3.3×10^{-8} (6.11)
	SH	1.7×10^{-2} (2.10)	2.4×10^{-5} (4.15)	3.4×10^{-8} (6.11)
three layers, interfaces 3 m away	P	6.3×10^{-3} (2.03)	8.7×10^{-7} (4.04)	2.4×10^{-9} (6.06)
	SV	1.0×10^{-2} (2.10)	5.8×10^{-6} (4.08)	5.2×10^{-8} (6.11)
	SH	1.3×10^{-2} (2.10)	3.0×10^{-6} (3.95)	9.3×10^{-8} (6.10)
three layers, interfaces against the voxels	P	4.8×10^{-3} (2.04)	2.4×10^{-6} (4.04)	4.1×10^{-9} (6.03)
	SV	8.5×10^{-3} (2.09)	7.1×10^{-6} (4.01)	5.0×10^{-8} (6.09)
	SH	1.4×10^{-2} (2.10)	5.8×10^{-6} (3.92)	8.5×10^{-8} (6.12)

to those of the whole space at the same frequency (Table 12): the background changes the field

the voxels must represent, not how well they represent it.

The two terms of the error at oblique incidence. The orders say that the scheme converges; the two-term analysis says what its error is. Both terms are tested at 20°, in the whole space and in the stratified backgrounds, for P, SV and SH waves.

The first-order term. The scheme is written as $(\mathsf{M}-\mathsf{K})\mathsf{c}=\mathsf{b}$ with the output $u=\mathsf{r}\,\mathsf{c}$, as in Eq. (53), so that $T_1^{\text{scheme}}=\mathsf{r}\,\mathsf{M}^{-1}\mathsf{b}$ with no differencing. The prediction is built from plane waves. The background field in the layer is a sum of six, P, SV and SH going down and going up, and the voxels radiate into six. For a pair made of an incident wave of wavevector $\mathbf{k}_{\text{in}}$ and a radiated wave of wavevector $\mathbf{k}_{\text{out}}$, a cubic voxel of half-side h whose field is held to total degree p gives the exact first-order term of that pair times

$$1+E_p(\mathbf{k}_{\text{in}},\mathbf{k}_{\text{out}})=\frac{\displaystyle\sum_{|a|\le p}\prod_{i=1}^{3}(2a_i+1)\,j_{a_i}(k_{\text{in},i}h)\,j_{a_i}(k_{\text{out},i}h)}{\displaystyle\prod_{i=1}^{3}j_0\big((k_{\text{in},i}-k_{\text{out},i})h\big)}, \tag{43}$$

the sum being over the Legendre products $P_{a_1}P_{a_2}P_{a_3}$ of the voxel's basis and j_n the spherical Bessel functions: the numerator is the projection of the incident wave onto the basis paired with that of the outgoing wave, the denominator the exact pairing. Its leading departure from one is $-d^2\,\mathbf{k}_{\text{in}}\cdot\mathbf{k}_{\text{out}}/12$ for $p=0$, with $d=2h$, and along one axis it is the factor of Eq. (19). The predicted first-order term is the sum over the thirty-six pairs of the exact term of each pair, an integral of two exponentials over the voxel, times its factor, the radiated waves then being carried to the observer by the reverberation of Eq. (41) exactly. On one and two planes of voxels of degree 0, 1 and 2, at $\omega=3000\,\text{rad}\,\text{s}^{-1}$, the scheme's first-order term equals this prediction to 2×10^{-15} in all twelve combinations of background and wave type. The first-order error is therefore known in closed form at oblique incidence, in a stratified background, for every wave type.

The second-order term. For the law of Eq. (24) the contrast must vary within the voxels. Each plane of voxels is given the smooth profile of §8 in depth, projected to Legendre degree $r=p$, and the coupling is that of Eq. (21) with the reverberation added. The second-order term of the scheme is $T_2^{\text{scheme}}=\mathsf{r}\,\mathsf{M}^{-1}\mathsf{K}\,\mathsf{M}^{-1}\mathsf{b}$. The exact T_2 needs the exact stack with a layer whose properties vary continuously in depth, at oblique incidence, for which a recursion over homogeneous layers is not exact. It is computed by two independent algorithms, invariant imbedding [50] and the propagator of the six-component state, which agree to 10^{-12} in all twelve cases (above). T_2 then follows from Eq. (54) at $\epsilon=0.1$, where halving ϵ changes it by 2×10^{-8}. The ratio of the relative error of T_2 to the relative projection error $\mathscr{E}_p$ is in Table 14. On four planes the law holds to 3% for $p=0$ and 1 and to 6% for $p=2$ at $k_SD=0.2$, and to 0.6% for all three degrees at $k_SD=0.067$, in every background and for every wave type: the departure falls about ninefold as the frequency falls threefold, as $(kD)^2$. On two planes the ratios for $p=0$ and 1 lie within 0.03 of those in the table; for $p=2$ they are lower, 0.89 to 0.99 at $k_SD=0.2$ and 0.988 to 0.999 at $k_SD=0.067$. The law is loosest for the highest degree at the highest

Table 14. The relative error of the second-order term over the relative projection error of the medium, at 20°, on $n = 4$ planes with $r = p$; the law of Eq. (24) predicts one in the long-wave limit.

		$k_S D = 0.2$			$k_S D = 0.067$		
background	wave	$p = 0$	$p = 1$	$p = 2$	$p = 0$	$p = 1$	$p = 2$
whole space	P	1.000	0.998	1.011	1.000	1.000	1.000
	SV	0.998	0.992	1.032	1.000	0.999	1.004
	SH	0.999	0.992	1.031	1.000	0.999	1.003
two half-spaces, interface 3 m below	P	1.010	1.009	1.019	1.001	1.001	1.001
	SV	1.012	1.008	1.046	1.001	1.001	1.005
	SH	1.027	1.025	1.059	1.003	1.003	1.006
three layers, interfaces 3 m away	P	1.004	1.003	1.013	1.000	1.000	1.000
	SV	0.979	0.973	1.002	0.997	0.996	1.000
	SH	1.007	1.004	1.035	1.001	1.001	1.004
three layers, interfaces against the voxels	P	1.002	1.001	1.011	1.000	1.000	1.000
	SV	0.995	0.988	1.018	0.999	0.999	1.002
	SH	1.004	1.002	1.032	1.000	1.000	1.003

frequency, because the projection error of degree 2 is so small (1.6×10^{-5} on four planes) that the departure from the long-wave limit is a larger share of it. Both terms of the error are thus known in advance at oblique incidence and in a stratified background, as they are at normal incidence in the whole space (`scripts/measure_oblique_two_term.py`).

Two limits stand. The oblique construction is implemented once, and its checks are the exact solutions. And the voxels here fill their plane, so that the image of a voxel in an interface it touches is the image of a whole plane; a body of finite lateral extent against an interface, where that image is singular, has not been run.

D The sphere: near field, an impedance march and a sharp surface

The near field. A receiver is seldom at infinity, and the scheme does not require it to be. The scattered field at a point $\mathbf{x}$ outside the voxels is the propagator applied to the source that each voxel carries, the contrast times the field found in it:

$$\psi^{\mathrm{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}')\, \boldsymbol{\Delta}(\mathbf{x}')\, e_j(\mathbf{x}')\, \mathrm{d}\mathbf{x}', \tag{44}$$

with $\mathbf{K}$ the nine-component Green's tensor of §2, the same one that couples the voxels in the solve, $\boldsymbol{\Delta}$ the contrast operator of Eq. (5) and e_j the polynomial internal state of voxel j. Its first three rows are the displacement and its last six the strain. Nothing is dropped: the terms in $1/r^2$ and $1/r^3$ are in $\mathbf{K}$. The integrand's source is a polynomial in each voxel and is integrated by a tensor Gauss rule with six nodes a side; the rule's own error is measured by repeating the readout with four nodes, and from 8 voxels across the two differ by less than 4×10^{-6} at $1.25a$ and by less than 10^{-10} at $2a$. The observers are the nine angles of Fig. 5(b) on spheres of radius $1.25a$, $2a$ and $10a$, where $k_S r$ runs from 0.6 to 10; the reference is the exact partial-wave displacement

at the same points; and the error is the measure of the far-field test. Table 15 gives it for the mean-only voxel and for the first-moment voxel (Appendix E). The orders of the far field hold in

Table 15. The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

		mean only			first moment		
	n_{sub}	$1.25a$	$2a$	$10a$	$1.25a$	$2a$	$10a$
$k_S a = 0.5$							
	4	3.3×10^{-2}	1.2×10^{-2}	5.3×10^{-3}	1.4×10^{-2}	1.6×10^{-3}	6.3×10^{-4}
	6	1.4×10^{-2}	5.2×10^{-3}	1.9×10^{-3}	1.2×10^{-3}	3.8×10^{-4}	1.9×10^{-4}
	8	8.0×10^{-3}	2.9×10^{-3}	1.1×10^{-3}	4.7×10^{-4}	1.4×10^{-4}	7.3×10^{-5}
	12	3.5×10^{-3}	1.2×10^{-3}	4.7×10^{-4}	7.8×10^{-5}	3.1×10^{-5}	1.7×10^{-5}
	16	2.0×10^{-3}	6.7×10^{-4}	2.6×10^{-4}	2.5×10^{-5}	1.0×10^{-5}	5.7×10^{-6}
order, 12 → 16		2.03	2.06	2.02	3.95	3.85	3.77
$k_S a = 1.0$							
	4	4.8×10^{-2}	3.5×10^{-2}	3.3×10^{-2}	9.9×10^{-3}	1.3×10^{-3}	6.0×10^{-4}
	6	2.0×10^{-2}	1.4×10^{-2}	1.4×10^{-2}	9.3×10^{-4}	3.0×10^{-4}	2.0×10^{-4}
	8	1.2×10^{-2}	8.1×10^{-3}	7.8×10^{-3}	3.5×10^{-4}	1.1×10^{-4}	7.4×10^{-5}
	12	5.2×10^{-3}	3.6×10^{-3}	3.5×10^{-3}	6.0×10^{-5}	2.5×10^{-5}	1.7×10^{-5}
	16	2.9×10^{-3}	2.0×10^{-3}	1.9×10^{-3}	1.9×10^{-5}	8.3×10^{-6}	5.7×10^{-6}
order, 12 → 16		2.01	2.01	2.00	3.95	3.84	3.78

the near field: second for the mean-only voxel at every distance and both frequencies, and 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the far field at this contrast. A quarter of a radius from the surface, with 16 voxels across, the first-moment voxel's error is 2.5×10^{-5} and 1.9×10^{-5}, 78 and 153 times below the mean-only voxel's. The error grows towards the body, by a factor of 1.5 to 7 between $10a$ and $1.25a$, because the near field carries the higher multipoles of the source, which a given grid resolves less well; the order does not change. On the coarsest grid a voxel is as wide as the graded shell and twice as wide as the gap to the nearest observer, and the sequence is not yet asymptotic. Read at these distances with the asymptotic formula, the same solutions would be in error by 48% to 85%: the far-field readout is for the far field (`scripts/measure_graded_sphere_near_field.py`; the readout is `graded_field` of the package, tested against the propagator of a point source, against the gradient of its own displacement, and against the far-field readout at great distance).

The voxels against an impedance march. A second route to the same body is invariant imbedding. The impedance of §8, Eq. (42), is marched upwards through the depth range of the sphere with the lateral variation of the medium carried inside the march, on a laterally periodic grid, and a second sweep carries the field down [50]; the march returns the reflected plane waves on a plane above the body and the transmitted ones on a plane below. It is run here on the graded sphere and set beside the voxels, to ask how many voxels a given accuracy costs against an established alternative. Four things are fixed so that the comparison is of like with like.

1. *The planes.* Both methods are read on the planes $z = -1.25a$ and $z = +1.25a$. On the tangent planes $z = \pm a$ themselves the observers lie on the faces of the voxels at the poles,

Table 16. The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

		$k_S a = 0.5$		$k_S a = 1.0$	
n_{sub}	unknowns	error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

where the Gauss rule of the voxel readout (Eq. (44)) cannot integrate the singular kernel: its change from four to six nodes is 2% to 10% there, more than the scheme's error, and below 10^{-4} a quarter of a radius away. The march's amplitudes are carried across the same homogeneous gap by each order's own phase, which is exact.

2. *The periodic array.* The march needs a laterally periodic grid, so it solves a square array of spheres. Its answer is scored against the exact solution of one sphere summed over the array, by Poisson summation, which is exact but for scattering between the spheres. That part, and the lateral resolution, are errors of the method as used on a single body, and are counted.

3. *The error.* For the march, the norm of the difference over the plane-wave orders of the grid, relative to the norm of the exact scattered orders; for the voxels, the norm of the difference of the scattered displacement over a patch of side $5a$ on each plane, relative to the norm of the exact one. The two are the same quantity in wavenumber and in space, for the array and for the single body.

4. *The refinement.* For the march, the lateral grid, with the depth step refined until the error no longer changes; for the voxels, the grid and the degree.

Table 17 gives the result. The march converges quickly in the lateral grid, because the body is smooth, and with far fewer unknowns than any voxel scheme: at 20×20 it is at 7×10^{-4} with 1200 unknowns on a plane. The mean-only voxel does not reach that with sixteen voxels across and twenty times the unknowns; on a sharp sphere, where it also carries a staircase, it fares worse. The first-moment voxel reaches the march's accuracy with six voxels across, and below about 2×10^{-4} it continues at close to fourth order where the march does not: in transmission the march's error at a period of $5a$ stops near 5×10^{-4}, which is the scattering between the spheres of its array, since lengthening the period lowers it and refining the grid does not. The costs are of different kinds. The march's unknowns are few but its operator is a dense matrix on the plane, marched through the depth, and a finer lateral grid requires a smaller depth step (with eight steps the 20×20 march diverged); the voxel schemes have many unknowns and a product done by fast Fourier transform. In the timings of these runs, taken on one machine with other work in progress and therefore only indicative, the first-moment voxel at six across was

Table 17. The graded sphere at $k_S a = 0.5$ read on the planes $z = \mp 1.25a$: relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is $5a$ unless stated.

method	setting	unknowns	reflection	transmission
impedance march	12×12, 16 steps	432	1.2×10^{-2}	9.9×10^{-3}
	16×16, 32 steps	768	4.3×10^{-3}	3.6×10^{-3}
	20×20, 32 steps	1200	7.6×10^{-4}	7.3×10^{-4}
	28×28, 48 steps	2352	2.5×10^{-4}	5.6×10^{-4}
	30×30, period $7.5a$	2700	7.2×10^{-4}	4.7×10^{-4}
mean-only voxel	8 across	3672	5.0×10^{-3}	7.9×10^{-3}
	16 across	24 984	1.2×10^{-3}	2.0×10^{-3}
first-moment voxel	6 across	7488	3.3×10^{-4}	5.2×10^{-4}
	8 across	14 688	1.4×10^{-4}	2.2×10^{-4}
	16 across	99 936	1.1×10^{-5}	1.8×10^{-5}
second-moment voxel	8 across	36 720	1.4×10^{-5}	2.3×10^{-5}
	10 across	69 120	2.6×10^{-6}	4.2×10^{-6}

solved in about a sixth of the time of the 20×20 march. The march has what the comparison does not use: it returns the reflection and transmission for every incident plane wave of the grid at once, and a laterally periodic or laterally extended medium is its natural problem and not a cost. The comparison is at one frequency (`scripts/measure_graded_sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

A sphere with a sharp surface. The homogeneous sphere is the familiar test, and its voxelisation changes with every grid. Table 18 shows its error for grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_S a = 0.5$. The error tracks the staircase's volume, which is 5%, 17% and 4% too large on these grids. Rescaled by the ratio of the sphere's volume to the staircase's, it falls below 1% at $k_S a = 0.5$ and to at most 3% at $k_S a = 1.0$, but it still does not converge: what remains is the staircase's shape. A refinement study of a sharp sphere measures its staircase; the smooth sphere, like the layer, measures the scheme.

Table 18. A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere's volume to the staircase's. It does not converge with refinement, because the staircase changes with the grid.

		$k_S a = 0.5$		$k_S a = 1.0$	
n_{sub}	voxels	raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

E The three-dimensional voxel solver

The three-dimensional results of Part II use the Legendre voxels of §7 on a cubic grid, with the following construction (`cubic_scattering/graded_voxel`).

The voxel. Each cell carries the nine-component state of §2, displacement and engineering strain, with the strain an *independent* unknown and not the gradient of the displacement. On the exact layer the difference is measured: with Legendre degree p an independent strain converges at order $2p+2$, while a strain derived from each cell's displacement converges at order 2, 2 and 4 for $p = 1$, 2 and 3, and adding the jumps of the displacement across the faces between cells as sources does not recover the order (`scripts/measure_derived_vs_independent_strain.py`). The state is expanded in the Legendre products of degree at most p in the cell's local coordinates $\boldsymbol{\xi} \in [-1,1]^3$, and the contrast operator $\boldsymbol{\Delta}$ in those of degree at most r, its coefficients the L^2 projection of the medium; $p = r = 1$ gives the first-moment voxel of 36 weights. The source $\boldsymbol{\Delta}\psi$ is then a polynomial of degree $p+r$, expanded in monomials m_c. The Lippmann–Schwinger equation is tested with each basis function over each cell (Galerkin), and cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m}\int_{V_n} L_a(\boldsymbol{\xi})\,\mathbf{P}(\mathbf{x}-\mathbf{x}')\,m_c(\boldsymbol{\xi}')\,\mathrm{d}^3x\,\mathrm{d}^3x', \qquad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \tag{45}$$

with $\mathbf{P}$ the 9×9 point propagator of §2.

One integral in place of two. With $\mathbf{s} = \mathbf{u} - \mathbf{u}'$ the difference of the two local positions, Eq. (45) becomes $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s})\,\mathbf{P}(\mathbf{R}+\mathbf{s})\,\mathrm{d}^3s$ over $\mathbf{s} \in [-2h, 2h]^3$, where W_{ac} is a product over the three axes of one-dimensional cross-correlations of the test and source polynomials: continuous, polynomial on $[-2h, 0]$ and $[0, 2h]$, with kinks at 0 and $\pm 2h$. For cells that do not touch, the kernel is smooth on each of the eight pieces, and a three-dimensional Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional quadrature of Eq. (45) to 10^{-9} one cell beyond contact.

The self cell and its 26 neighbours. The static part of $\mathbf{P}$ is a sum of derivatives of r^{-1} and r, distributions at the origin. Derivatives are moved onto W until the kernel is at most as singular as $1/r$, and the kinks of W then carry plane δ functions; every piece is a box or a rectangle with its corners on the breakpoints, so the singular point is either a vertex of a piece, integrated with Duffy's pyramids, or at least $2h$ away. The δ function of the inclusion problem is carried by those plane δ functions and is not added by hand. The construction reproduces the sum rule $\sum_p \int_V\int_V \partial_p\partial_p|\mathbf{x}-\mathbf{x}'|^{-1}\,\mathrm{d}^3x\,\mathrm{d}^3x' = -4\pi V$, the biharmonic rule for r and the cube's Coulomb self-energy to 10^{-11}, and every block equals the sum of the 64 blocks between the two cells' half-size sub-cells, with the polynomials re-expressed on them, to 2×10^{-15}. The dynamic remainder of $\mathbf{P}$ is at most weakly singular and is integrated by the Gauss rule of the pieces.

Symmetry, and the solve. A block depends only on the offset $\mathbf{o}$ between the two cells. The background is isotropic and the cube is mapped to itself by its 48 signed permutations $\mathbf{Q}$, under which $\mathbf{P}(\mathbf{Qr}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^{\mathsf{T}}$, with $\mathbf{S} = \mathrm{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$ and $\mathbf{R}_6$ the representation of $\mathbf{e} \mapsto \mathbf{QeQ}^{\mathsf{T}}$ on Voigt vectors. With the test and source polynomials transforming by matrices $\mathbf{T}$ and $\mathbf{U}$, $\mathbf{K}_{ac}(\mathbf{Qo}) = \sum_{b,d} T_{ab}U_{cd}\,\mathbf{S}\mathbf{K}_{bd}(\mathbf{o})\mathbf{S}^{\mathsf{T}}$, so one block is computed for each orbit of offsets and the rest follow exactly. The coupling is then a convolution over the grid, applied by the fast Fourier

transform on a $(2n-1)^3$ circular embedding, and the system is solved by GMRES preconditioned with each cell's own block. Where both can be run, the solution equals that of a dense solve to 10^{-10} relative (`tests/test_graded_voxel_fft.py`). Every cell that overlaps the body is kept, each with the L^2 projection of its contrast by a six-point Gauss rule per axis; keeping only the cells whose centre lies inside the body drops the contrast of the partly filled cells, and the mean-only voxel then converges at order one instead of two. The far field is read at 5×10^8 radii, where the near-field terms that the asymptotic form omits are negligible.

The first-moment voxel on the graded sphere. The voxel with $p = r = 1$ is applied to the graded sphere of §10. To isolate the discretisation error from the rescattering the grid cannot yet resolve, the contrast is first scaled by 10^{-6}, and the exact sphere is solved at the same scaled contrast, so that no Born approximation enters the reference: the relative error is then that of the scheme's first-order term, in which the field is the incident one carried across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the error is 7.5×10^{-5}, 5.0×10^{-6} and 1.6×10^{-6} at $k_S a = 0.5$, and 2.3×10^{-4}, 9.1×10^{-5} and 2.8×10^{-5} at $k_S a = 1$. The apparent orders between 6 and 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a voxel at $n = 4$ being as wide as the graded shell (`scripts/pilot_graded_voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16 voxels across, where the error is 4.9×10^{-6} at $k_S a = 0.5$ and 5.9×10^{-6} at $k_S a = 1$, and the apparent order approaches four at a rate set by the number of voxels across the graded shell.

F The scattering series of the layer, and the projection-error law

The scattering series. Multiply the whole contrast, $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$ together, by a number s, and let $u(s)$ be the scattered displacement at an observer at depth z_o outside the layer. It vanishes at $s = 0$ and is analytic in s there, and its expansion is the Born series,

$$u(s) = sT_1 + s^2T_2 + s^3T_3 + \dots, \tag{46}$$

so that T_n is the part of the field at the full contrast, $s = 1$, that has been scattered n times.

The first two terms of the exact layer. They are integrals over the layer. With the incident state $w^0 = (u^0, \varepsilon^0)$, the contrast $\mathbf{\Delta}_1 = \mathrm{diag}(\omega^2\Delta\rho, \Delta M)$ with profile $f(z)$, and the kernel $\mathbf{K}$ of Eq. (18), whose first row $(g, g')(z_o - z)$ gives the displacement at the observer,

$$T_1 = \int_0^D (g,\, g')(z_o - z)\; \mathbf{\Delta}_1 f(z)\, w^0(z)\, \mathrm{d}z, \tag{47}$$

$$w^1(z) = \int_0^D \mathbf{K}(z - z')\, \mathbf{\Delta}_1 f(z')\, w^0(z')\, \mathrm{d}z', \qquad T_2 = \int_0^D (g,\, g')(z_o - z)\; \mathbf{\Delta}_1 f(z)\, w^1(z)\, \mathrm{d}z. \tag{48}$$

Here w^1 is the state scattered once, inside the layer; its strain includes the local term of the kernel, $-f(z)\,\Delta M\,\varepsilon^0(z)/M_P$. Each further term repeats the step, $w^n(z) = \int_0^D \mathbf{K}(z - z')\, \mathbf{\Delta}_1 f(z')\, w^{n-1}(z')\, \mathrm{d}z'$. For a plane force at z_s above the layer, $w^0(z) = c\, \mathrm{e}^{\mathrm{i}k(z-z_s)}(1, \mathrm{i}k)$ with $c = \mathrm{i}/(2M_P k)$, every integrand

is an exponential times the profile. Writing $a = \omega^2\Delta\rho$ and $b = \Delta M$, and carrying out the products of Eqs. (47) and (48), the terms seen in reflection ($z_o < 0$) and in transmission ($z_o > D$) are, for any profile,

$$T_1^R = c^2 \mathrm{e}^{-\mathrm{i}k(z_o+z_s)}\big(a + bk^2\big)\int_0^D f(z)\,\mathrm{e}^{2\mathrm{i}kz}\,\mathrm{d}z, \qquad T_1^T = c^2 \mathrm{e}^{\mathrm{i}k(z_o-z_s)}\big(a - bk^2\big)\int_0^D f(z)\,\mathrm{d}z, \tag{49}$$

$$\begin{aligned} T_2^R &= c^2 \mathrm{e}^{-\mathrm{i}k(z_o+z_s)}\Big[\,c\,(a^2 - b^2k^4)\int_0^D\!\!\int_0^D f(z)f(z')\,\mathrm{e}^{\mathrm{i}k(z+z'+|z-z'|)}\,\mathrm{d}z'\,\mathrm{d}z - \frac{b^2k^2}{M_P}\int_0^D f(z)^2\,\mathrm{e}^{2\mathrm{i}kz}\,\mathrm{d}z\Big], \\ T_2^T &= c^2 \mathrm{e}^{\mathrm{i}k(z_o-z_s)}\Big[\,c\,\big(a - bk^2\big)^2\int_0^D\!\!\int_0^z f(z)f(z')\,\mathrm{d}z'\,\mathrm{d}z + c\,\big(a + bk^2\big)^2\int_0^D\!\!\int_z^D f(z)f(z')\,\mathrm{e}^{2\mathrm{i}k(z'-z)}\,\mathrm{d}z'\,\mathrm{d}z \\ &\qquad + \frac{b^2k^2}{M_P}\int_0^D f(z)^2\,\mathrm{d}z\Big]. \end{aligned} \tag{50}$$

The first-order reflection is the Fourier transform of the profile at the wavenumber $2k$, and the first-order transmission its integral. In each second-order term the last integral is the local term of the kernel, in which both scatterings are at one point and the profile appears squared; the others are the two scatterings at different depths. In transmission the wave scattered twice forwards carries the contrast of velocity twice, $(a - bk^2)^2$, and the wave scattered backwards and then forwards again the contrast of impedance twice, $(a + bk^2)^2$. For the uniform layer ($f = 1$) the integrals give

$$T_1^R = c^2 \mathrm{e}^{-\mathrm{i}k(z_o+z_s)}\,\big(a + bk^2\big)\,\frac{\mathrm{e}^{2\mathrm{i}kD} - 1}{2\mathrm{i}k}, \qquad T_1^T = c^2 \mathrm{e}^{\mathrm{i}k(z_o-z_s)}\,\big(a - bk^2\big)\,D, \tag{51}$$

$$\begin{aligned} T_2^R &= \frac{c^2 \mathrm{e}^{-\mathrm{i}k(z_o+z_s)}}{4k^3M_P}\Big[-\mathrm{i}\big(a^2 + b^2k^4\big) + \mathrm{e}^{2\mathrm{i}kD}\big(a^2(\mathrm{i} + 2kD) + b^2k^4(\mathrm{i} - 2kD)\big)\Big], \\ T_2^T &= \frac{\mathrm{i}\,c^2 \mathrm{e}^{\mathrm{i}k(z_o-z_s)}}{8k^3M_P}\Big[a^2\big(1 - \mathrm{e}^{2\mathrm{i}kD} + 2kD(\mathrm{i} + kD)\big) - 2abk^2\big(\mathrm{e}^{2\mathrm{i}kD} - 1 + 2kD(kD - \mathrm{i})\big) \\ &\qquad + b^2k^4\big(1 - \mathrm{e}^{2\mathrm{i}kD} + 2kD(kD - 3\mathrm{i})\big)\Big]. \end{aligned} \tag{52}$$

The reflection sees the contrast of impedance, $a + bk^2 = k^2M_P(\Delta\rho/\rho + \Delta M/M_P)$, and the transmission that of velocity, $a - bk^2$. As $kD \to 0$, $T_2^R \to -c^2\mathrm{e}^{-\mathrm{i}k(z_o+z_s)}\,b^2k^2D/M_P$: the local term alone, which is the starting point of the argument at the end of this appendix. For the smooth profile the integrals of Eqs. (49) and (50) are evaluated in closed form symbolically, and by quadrature they return the same values to 10^{-16}. Equations (51) and (52) equal, identically, the first two Taylor coefficients in s of the exact reflection and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At $\omega = 300\,\mathrm{rad\,s^{-1}}$, $|T_2|/|T_1|$ is 0.039 in reflection and 0.17 in transmission for the uniform layer.

The first two terms of the scheme. Write Eq. (18) as $(\mathsf{M} - \mathsf{K})\,\mathsf{c} = \mathsf{b}$, with M the diagonal matrix of the $\int_i \phi_a^2\,\mathrm{d}z$, K the matrix of the coupling blocks times $\mathbf{\Delta}_1$, and b the moments of the incident field, and let r be the row that gives the scattered displacement at the observer from the

coefficients, $u = \mathsf{r}\,\mathsf{c}$. Scaling the contrast by s scales K and r by s, so $u(s) = s\,\mathsf{r}(\mathsf{M} - s\mathsf{K})^{-1}\mathsf{b}$ and

$$T_1^{\text{scheme}} = \mathsf{r}\,\mathsf{M}^{-1}\mathsf{b}, \qquad T_2^{\text{scheme}} = \mathsf{r}\,\mathsf{M}^{-1}\mathsf{K}\,\mathsf{M}^{-1}\mathsf{b}. \tag{53}$$

$\mathsf{M}^{-1}\mathsf{b}$ is the projection of the incident field onto each cell's functions: the first term needs no solve. Splitting K into its blocks within a cell and its blocks between cells splits T_2^{scheme} into the part scattered twice in one cell and the part scattered in two; the same split of the inner integral of Eq. (48) splits the exact T_2.

A check by differences. The terms can also be read from solutions at four scaled contrasts, $s = \pm\epsilon$ and $\pm 2\epsilon$, with no knowledge of the formulas above. The odd and even parts of u in s separate the odd orders from the even, and the two step sizes remove the next term of each (the five-point central differences for $u'(0)$ and $\frac{1}{2}u''(0)$):

$$\begin{aligned} T_1 &= \frac{8\big[u(\epsilon) - u(-\epsilon)\big] - \big[u(2\epsilon) - u(-2\epsilon)\big]}{12\,\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16\big[u(\epsilon) + u(-\epsilon)\big] - \big[u(2\epsilon) + u(-2\epsilon)\big]}{24\,\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \tag{54}$$

A negative s is a medium slower and lighter than the background by the same fractions, as legitimate a medium as the other. With $\epsilon = 0.01$ the remainders $4\epsilon^4 T_5/T_1$ and $4\epsilon^4 T_6/T_2$ are below 2×10^{-12} for this layer, and the formulas applied to the exact layer return Eqs. (51) and (52) less exactly those remainders, to 10^{-17}. The measurements of §8 use the differences for the scheme and the exact layer alike, each in its own arithmetic, and they agree with Eqs. (47), (48) and (53).

Why the terms must be separated. The law of Eq. (24) is a statement about T_2. At the full contrast the terms of third and higher order are 7% of the second and carry errors of their own, and if every order above the first is taken together the ratio of Eq. (24) reads 0.90 to 0.94 where the second-order term alone gives 1.00 to 1.01. The shortfall is itself the single site's. In the long-wave limit the kernel is local, the exact strain is $\varepsilon_0/(1 - cf)$ with $c = -\Delta M/M_P$, and the scheme's is the Galerkin solution of $\varepsilon = \varepsilon_0 + cf\varepsilon$ in each cell on that cell's polynomials; the relative error of the term of order n is $(\int f^n\,\mathrm{d}z - \int(\Pi f)^n\,\mathrm{d}z)/\int f^n\,\mathrm{d}z$, 2.455 times $\mathscr{E}_q$ for the third term of this profile, which has the opposite sign to the second. At $\omega = 30\,\mathrm{rad\,s^{-1}}$ the whole nonlinear part has 0.896 times $\mathscr{E}_q$, and the cell-by-cell Galerkin solution reproduces it to four digits with c from the contrast (`scripts/measure_layer_t3_law.py`).

Equation (24) is exact in the long-wave limit, for a reason that shows what each basis is for. The strain–strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$, is local as $kD \to 0$, and the incident strain ε_0 is then uniform across the layer, so the strain scattered once is $-(\Delta M/M_P)\,f(z)\,\varepsilon_0$: it is the profile itself. A cell holds the contrast as $\Pi_r f$ and the field to degree p, so it holds the once-scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order term pairs the contrast with that strain:

$$T_2 \;\propto\; \int f \cdot f\,\mathrm{d}z = \|f\|^2, \qquad T_2^{\text{scheme}} \;\propto\; \int (\Pi_r f)(\Pi_q f)\,\mathrm{d}z = \|\Pi_q f\|^2, \tag{55}$$

the last because $\Pi_q f$ lies in the range of Π_r. The relative error is $1 - \|\Pi_q f\|^2/\|f\|^2 = \mathscr{E}_q$. Because the kernel is local in this limit, the second scattering takes place in the cell of the first: T_2 is the second-order term of the single-site T-matrix, and Eq. (24) is a statement about the single site. The T-matrix sums the rescattering within a cell exactly for the cell as it is represented, a contrast $\Pi_r f$ and a field of degree p; its error is that the cell is not the medium. The split is measured directly (`scripts/measure_layer_bases.py`): at $\omega = 300\,\mathrm{rad\,s^{-1}}$ on 4, 8 and 16 cells, T_2 of the scheme and of the exact layer are each divided into the part whose two scatterings lie in one cell and the part between different cells. For all nine pairs the error of the same-cell part is 0.998 to 1.013 times $|T_2|\,\mathscr{E}_{\min(p,r)}$, and that of the part between cells at most 0.002 times it. The coupling between cells contributes only the departure from the limit. Measured, the ratio of the two sides of Eq. (24) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2}, 4.3×10^{-3}, 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24$, 0.12, 0.04 and 0.012: as $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (55) is verified symbolically for all nine pairs, and the projection errors are computed exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each halving of the cells.

G Relation to other methods

Relation to the coupled-dipole method. The same partition appears in the coupled-dipole method as the choice of polarisability. The Clausius–Mossotti relation, the lattice dispersion relation [13], the polarisability taken from the exact static solution [14] and the integrated Green tensor [15] all adjust the single site so that the lattice reproduces the continuum. For space-filling cubes the adjustment is neither needed nor a free parameter: the tiling identity fixes the local field exactly, the self term of the collocation closure is the one that makes it cancel, and what is left is a closed-form error rather than an empirical one. That the method is a collocation of the volume integral equation was shown for the electromagnetic case by Lakhtakia [11]; Eq. (15) is the same statement for the elastic lattice with the self term evaluated over the finite cube, and it is what makes the error a midpoint-rule error. The filtered coupled dipoles of Piller and Martin [16] and Yurkin et al. [17] band-limit the source to the grid; the source-cell average does the complementary thing exactly, its form factor vanishing on the reciprocal lattice, and the spectral construction of Eq. (40) is the periodic coupled-dipole method of Chaumet et al. [18] with the Galerkin double average in place of the point dipole. The second-order convergence reported for the DDA by Yurkin et al. [4] corresponds to the uniform-field voxel's, and the opposite-signed constants of Eq. (16) explain why a polarisability correction can improve one observable only at the expense of another, as the package's own form-factor corrections do in reflection and transmission. Going beyond second order requires a richer internal field, which is the point of higher-order bases for volume integral equations [2, 35]; the first-moment voxel is the lowest such basis, and in its spectral form the Galerkin double average makes the lattice sum converge without Ewald splitting.

Relation to the spectral element method. The voxel of degree p is close kin to the spectral element [52], which was taken up for seismic wave propagation in the 1990s, is set out for

three-dimensional models by Komatitsch and Tromp [53], and is now a standard tool for it. Both divide the medium into hexahedra, represent the field in each by polynomials of high degree, and determine the coefficients by a Galerkin condition on a weak form. The spectral element uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of the element; the voxel uses the Legendre polynomials themselves. Four of the present results have their counterparts there.

1. *A diagonal mass matrix.* The spectral element obtains one by integrating at the Gauss–Lobatto–Legendre points, at which each interpolant vanishes but one [53]. That rule is not exact for the product of two interpolants, so that the diagonal form is reached, in their words, by a process of subintegration, as in the finite-element methods that lump the mass. Here the Gram matrix M of Eq. (53) is diagonal with no approximation, because the Legendre functions are orthogonal on the cell; this is why the first-order term needs no solve. The two routes serve different ends: theirs an explicit march in time, ours the terms of the scattering series.

2. *The doubled order.* A scattered amplitude is a functional of the field, and a Galerkin method gives a functional to twice the order of the field itself. The order $2p + 2$ is this doubling: the leading error of Eq. (19) is the product of the part of the incident wave that the cell's polynomials miss and the part of the outgoing wave that they miss, each of order h^{p+1}.

3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [53]. The rule $\min(2p + 2, 2r + 2)$ says why that is the right choice, and what a coarser representation of the medium would cost.

4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [53]. The stratified layer of §8 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [53]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a coupling between every pair of voxels, and a background restricted to one whose Green's tensor is known. It differs also from the boundary integral methods, which are restricted to a finite number of homogeneous regions [53]: a volume integral equation admits a medium that varies within the scatterer. The voxel scheme is therefore a candidate where the heterogeneity is localised in a layered medium, and not a replacement for

the spectral element in a model that is heterogeneous throughout. A spectral-element mesh is designed by a rule of resolution, about five grid points to the shortest wavelength at polynomial degrees between five and ten [53]. The error analysis of this paper is the counterpart of that rule for the integral formulation: it gives the size and the degree of the voxels for a required accuracy in closed form, and it adds a term, the projection error of the medium.

H The cost of higher order, and what the error analysis offers a fast solver

The cost of fourth order. A first-moment voxel carries four times as many unknowns as a uniform one (36 against 9 in three dimensions). The closed forms show what that buys. To reach an error ε in reflection, the uniform voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\,\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$ that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction. In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the number of unknowns by about 5×10^4 (these are the Born-order constants; at the contrast studied the first-moment voxel's measured errors lie within 5% of them, while the uniform voxel's transmission constant grows with the contrast, so the comparison is if anything conservative). The stratified model of §8 measures it directly: 347 planes of uniform-field voxels against 8 first-moment voxels for the same accuracy. Higher moments continue the trend in one dimension: by Eq. (20) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already a quarter of a wavelength. In three dimensions, with the Legendre products of total degree at most p, a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for the first-moment voxel. The voxel of degree 2 converges at sixth order in the layer at oblique incidence (Table 12); that of degree 3 has not been built in three dimensions. The gain is largest precisely where it matters: when the model varies on scales that are coarse compared with the accuracy the wavefield must have.

What the error analysis offers a fast solver. A voxel scheme is dense, and for a large medium it is solved iteratively with a fast product. The adaptive integral method [54] and the precorrected fast Fourier transform [55] are the standard forms: the sources are projected onto a regular grid, in the first by matching their multipole moments and in the second by matching the potentials they produce at a set of test points, the interactions between distant cells are evaluated on that grid by fast Fourier transforms, and those between near cells are computed directly and corrected. Such a method has three accuracies to set, that of the discretisation, that of the projection, and the radius of the near zone, and they are usually set by trial. The results of this paper set them beforehand, in five ways.

1. *The target is known.* The discretisation error is known before the solve, in order and in size: the first-order factor of a cell, Eq. (19) and its form for any pair of wavenumbers (§8), and the projection error of the medium, Eq. (24). A fast product need be no more accurate than that, and it is known when it is accurate enough.

2. *The two errors do not mix.* The error of the scheme is the single site's: the coupling between

cells, averaged over the cells, is exact (§5). An approximation of the far coupling to a set tolerance therefore adds its own error to the discretisation's and does not alter it, so the two can be budgeted separately.

3. *For cubes on a lattice there is nothing to project.* The cells are the regular grid. The coupling depends on the separation of the cells alone, so the product is a convolution, block by block in the polynomial degrees, and the fast Fourier transform applies to it exactly, as in Georgakis et al. [2] for the linear voxel. What the projection onto the grid approximates in general is here an identity.

4. *For cells of several sizes the moments are already in hand.* Where the cells differ in size or degree and are mapped to a uniform auxiliary grid, the quantities to be matched are the polynomial moments of a cell, which are the unknowns of the scheme; and the first-order factor says how many of them matter at a given kd.

5. *A stratified background splits as the method requires.* The Green's tensor of a plane-stratified background is the whole-space tensor, which is invariant under translation in three dimensions and carries all the singularity, plus a reverberation, Eq. (41), which is smooth, is invariant under lateral translation, and separates in depth into one-dimensional moments. The first is the convolution above; the second is a fast Fourier transform laterally and a product in depth, with no near-zone correction except where a voxel touches an interface. A splitting of the kernel into a term that carries the singularity and a nonsingular remainder is used to a related end by Sharma and Triverio [56], for surface integral equations in homogeneous and in layered backgrounds: there the singular term is independent of frequency, so that the direct near-zone integrals are computed once for a whole sweep in frequency. And because the orders and the error law are unchanged by the background (§8), the error budget of the whole space carries over to it.

No such solver is built here. These are consequences of the analysis, and they are to be tested.

I The evidence for each result, and where to reproduce it

Table 19 lists each result with the evidence for it. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, GradedContrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the stratified reference against Kennett's recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded, with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_sphere.py`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the independent implementation of the first-moment voxel is `scripts/`

`crosscheck_first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_graded_contrast.py`; the projection-error law and its split are measured by `scripts/measure_layer_bases.py`, in a stratified background by `scripts/measure_layer_stratified_background.py`, `scripts/measure_oblique_stratified_background.py` and `scripts/measure_oblique_two_term.py`, and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_degree.py`; the convergence figure is drawn by `scripts/plot_convergence_orders.py`; the near field of the sphere is measured by `scripts/measure_graded_sphere_near_field.py`; All are available at https://github.com/tmnestor/MultipleScatteringCalculations.

Table 19. The results of Part I and of the sphere of §10, and the evidence for each; those of Part II are in §14.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8}; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \to 0$	11× the continuum term at $m = 1$
Averaged coupling between planes = continuum× sinc	2.8×10^{-16}, 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Self term cancels, Eq. (15)	4×10^{-16}
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (16)	$n = 1$–16, to 10^{-4} of the error
First-moment voxel, Eq. (18)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (40)	order 4, P and S, 20° and 40°
Incident SV and SH, all schemes	orders 2 and 4; constants with k_S to 4×10^{-4}
First-moment error $-(kd)^4/720$, Eq. (19)	$n = 1$–8, to 10^{-4} of the error
Voxel of degree p: order $2p + 2$	6.00 ($p = 2$, three frequencies), 8.00 ($p = 3$)
Constant c_p, Eq. (20)	symbolic, $p = 0$–5, twice; chain to 4×10^{-7}
One plane of degree p = thin-layer series to D^{2p+2}	exact series, $p = 0$–3, two layers; $D^{3.00}$, $D^{5.00}$, $D^{7.00}$
Independent implementation, any degree	within round-off; orders 4.02, 6.02
Independent implementation of Eq. (40)	1.1×10^{-15}, 40 cases: P, SV, SH, stratified
Random stratified layer, §8	orders 2 and 4, P and S, normal and 20°
Stratified reference against Kennett's recursion	10^{-14}
Contrast varying within a cell, Eq. (22)	8 cases, linear and smooth profiles; closed forms tested one by one to 2×10^{-16}
Projection-error law, Eq. (24)	nine pairs (p, r) to 1.3%; departure as $(kD)^2$; pairing symbolic, nine pairs
Stratified background, Eq. (25), normal incidence	orders 2, 4, 6; first-order term to 5×10^{-16}; law to 10^{-4} at $kD = 0.012$
Stratified background at 20°, Eq. (41), Table 13	orders 2, 4, 6 in 36 cases; background reflection against Kennett to 5×10^{-15}
First-order term at 20°, Eq. (43)	36-pair prediction to 2×10^{-15}, 12 cases, $p = 0, 1, 2$
Projection-error law at 20°, Table 14	$p = 0, 1, 2$: to 6% at $k_S D = 0.2$, 0.6% at 0.067; graded reference by impedance march and by propagator, 10^{-12}
The error of T_2 is the single site's	same-cell part 0.998 to 1.013 of the law; between cells ≤ 0.002
Degree 2 at oblique incidence, Table 12	order 6, P, SV and SH, against Kennett's recursion
Smoothly graded sphere, §10	exact solution by two routes, 2×10^{-11}; orders 2.04 and 1.99, $n_{\text{sub}} = 6$ to 24
Near field of the sphere, Table 15	orders 2.0 and 3.8 to 3.95 at 1.25, 2 and 10 radii, two frequencies
Voxels against the impedance march, Table 17	same planes, one frequency; march's floor in transmission traced to its

Acknowledgments

This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [3].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow, organised as a Test Driven Hypothesis Acceptance cycle: each derivation, implementation or numerical claim was stated as a hypothesis together with a test designed to refute it (an independent route, an exact limit, a high-precision reference or a convergence study), and was accepted only when the test passed; a failed test returned the hypothesis for diagnosis, never the test for adjustment. The author designed the software originally, but only directed and validated recent improvements, which build on the author's original Fortran implementation (the `PhD_fortran_code` folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus 5.5) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author's. All AI-generated changes were subject to code review by the author and validated by automated tests and independent physical checks such as reciprocity, optical theorem, Hamiltonian symmetry, analytic limits and convergence.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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